

Modelling Financial Markets

Chapter 12: Options and the Volatility Surface



Daniel Traian PELE

Bucharest University of Economic Studies
IDA Institute Digital Assets
Blockchain Research Center
AI4EFin Artificial Intelligence for Energy Finance
Romanian Academy, Institute for Economic Forecasting
MSCA Digital Finance

What Do Option Prices Reveal About the Future Distribution? (1/2)

- An **option** gives the right, not the obligation, to buy (call) or sell (put) an asset at a fixed strike price
  - It pays only in some states: a bet on the **distribution**, not only on the direction; option prices reveal what the market expects about volatility and crashes
- Part I: pricing by replication and what a hedge leaves behind
  - Parity and static spanning; the tree and the Black–Scholes equation; the Greeks; discrete-hedging error and the gamma-weighted volatility P&L (profit and loss)
  - Implied volatility, the smile, and a full Bitcoin surface with its risk-neutral density

What Do Option Prices Reveal About the Future Distribution? (2/2)

- Part II: implied volatility as information
  - Variance swaps and the VIX (the Cboe index of 30-day S&P 500 implied volatility); the variance risk premium; is the VIX a good forecast? 0DTE options (zero days to expiry)
  - Case study: are option-implied bounds on the equity premium valid and tight? AI (artificial intelligence) for scientific discovery
- Outcome: estimate a surface and a density under no-arbitrage constraints, with bands, and test option-implied forecasts with the right standard errors

Chicago, 1973: Listed Options Begin



Trading floor of the Chicago Board of Trade in 2006; its members founded the Cboe in 1973

Photo: Richie Diesterheft (2006); CC BY 2.0; Wikimedia Commons

- The Chicago Board Options Exchange (Cboe) opened in 1973 as the first exchange for standardised, listed options MacKenzie & Millo (2003)
  - Before: options were negotiated one by one, over the counter
- The same year: the Black–Scholes and Merton pricing papers Black & Scholes (1973), Merton (1973)
- Traders soon used the formula on the floor: theory changed the market it described MacKenzie & Millo (2003)
- Textbooks
  - Franke, Härdle & Hafner (2019)
  - Härdle, Chen & Overbeck (2017), Gatheral (2006)

The Formula and Its Authors



Fischer Black (1938–1995)

Photo: Dalmatine at English Wikipedia; public domain; Wikimedia Commons



Myron Scholes (2008)

Photo: Nobellaureatesphotographer (2008); CC BY-SA 3.0; Wikimedia Commons



Robert C. Merton (2006)

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- Key idea: a portfolio of the stock and cash can **replicate** the option, so its price follows from no-arbitrage Black & Scholes (1973)
- Scholes and Merton received the 1997 Nobel Memorial Prize in Economic Sciences; Black had died in 1995

The Data: Volatility Indices and a Bitcoin Option Snapshot

Series	What it measures	Period	Owner
S&P 500	US equity index, daily close	1990–2026	S&P Dow Jones Indices
SPY	exchange-traded fund (ETF) on the S&P 500: open and close	1993–2026	State Street
VIX	30-day implied volatility of the S&P 500	1990–2026	Cboe
VIX9D, VIX3M	the same over 9 days and 3 months	VIX9D: 2011–2026; VIX3M: 2007–2026	Cboe
VVIX	implied volatility of the VIX	2006–2026	Cboe
Bitcoin options	all listed contracts, one dated snapshot	22 Sep 2026	Deribit
DVOL	30-day implied volatility of Bitcoin	2021–2026	Deribit

- Daily data up to 18 September 2026; the option snapshot was taken on 22 September 2026, 07:51 UTC (UTC: Coordinated Universal Time)
- Implied volatility is quoted in annualised %; "vol points" = percentage points of volatility

Notation: Prices, Expiry and Moneyness

- Prices and parameters
  - S_t: price of the underlying at time t; K: strike; T: expiry; r: continuously compounded risk-free rate; q: dividend yield
  - C, P: prices of a call and a put; $(x)^+ = \max(x, 0)$; σ_{imp}: implied volatility (Section 6)
- European call and put pay off $(S_T - K)^+$ and $(K - S_T)^+$ at expiry T; $\tau = T - t$ in years; American options can be exercised at any time up to T
  - Both pay-offs are convex in S_T, so an option's value increases with volatility (vega > 0, Section 4)

Moneyness and Context-dependent Symbols

- Forward $F = Se^{(r-q)\tau}$, log-moneyness $k = \ln(K/F)$; ATM (at the money): $k \approx 0$; OTM (out of the money): puts with $K < F$ and calls with $K > F$; ITM (in the money): the opposite
  - Most of this chapter works in k and in total implied variance $w = \sigma_{imp}^2 \tau$
- One letter, several roles (read them from the context): q is the dividend yield; $q(K)$ the risk-neutral density of S_T (Sections 2 and 7); q in the binomial tree the risk-neutral up-probability (Section 3); Q the pricing measure; $Q(K_i)$ the Cboe mid quote at strike K_i (Section 8)

Model-Free Bounds, Log Returns and Prerequisites

- Model-free bounds (no dividends): $\max(S - Ke^{-r\tau}, 0) \le C \le S$ and $\max(Ke^{-r\tau} - S, 0) \le P \le Ke^{-r\tau}$
  - With $r > 0$ an American call on a stock without dividends is never exercised early: it is worth at least $S - Ke^{-r\tau} > S - K$
- Moneyness is measured here against the forward F; the pay-off primer of Seminar 12 (Options and Their Pay-offs) and the smile chart of Section 6 use the spot S (OTM call: $K > S$)
  - The two references differ by the factor $F/S = e^{(r-q)\tau}$, close to 1 for short expiries: only strikes between S and F change class
- Log return $\ln(S_t/S_{t-1})$, not the simple return $(S_t - S_{t-1})/S_{t-1}$: from 100 to 110 the simple return is 10%, the log return $\ln 1.1 = 9.53\%$
  - Log returns add up over time and are Normal under Black–Scholes; volatility σ is their annualised standard deviation
- Assumed: Brownian motion, Itô's lemma and martingales (Chapter 11); regression with HAC (heteroskedasticity and autocorrelation consistent) standard errors (Chapter 8)

Put–Call Parity: What Does No Arbitrage Alone Imply?

- Two portfolios with the same pay-off at T (European options, same K and T, no dividends):
  - A: one call + cash $Ke^{-r\tau}$ (worth K at T) $\Rightarrow \max(S_T, K)$
  - B: one put + one share $\Rightarrow \max(K, S_T)$
- Same pay-off in every state, so the same price today, otherwise there is an arbitrage:
  - $C_t - P_t = S_t - Ke^{-r\tau}$ (with dividend yield q: $S_t e^{-q\tau}$ instead of S_t)
- No model for S is needed: parity uses only no-arbitrage Merton (1973)
- Consequence: a call and a put with the same K and T have the **same** implied volatility

Static Replication: Any Pay-Off from a Bond, a Forward and Options

- Carr–Madan spanning: any twice-differentiable pay-off $f(S_T)$ and any point F (Taylor with integral remainder; proof in the Appendix):
  - $f(S_T) = f(F) + f'(F)(S_T - F) + \int_0^F f''(K)\,(K - S_T)^+ dK + \int_F^\infty f''(K)\,(S_T - K)^+ dK$
  - f', f'': first and second derivatives of the pay-off; reading: a bond, a forward and $f''(K)\,dK$ puts below F and calls above F
  - Priced today (F the forward): $V_0 = e^{-r\tau} f(F) + \int_0^F f''(K)\,P(K)\,dK + \int_F^\infty f''(K)\,C(K)\,dK$
  - $P(K)$, $C(K)$: today's put and call prices at strike K, whatever the model for S Bakshi, Kapadia & Madan (2003)
- Why this price: each piece has a market price today (the bond $e^{-r\tau}$ per unit, the forward 0, the options $P(K)$ and $C(K)$); two positions with the same pay-off at T must cost the same, exactly as in put–call parity
  - No model for S and no pricing measure are needed for V_0; the measure Q of Section 3 is needed only to read option prices as expectations, as in the density bullet of the next slide

Static Replication: Three Results Used Later

- "Risk-neutral": a price is the discounted mean pay-off under the pricing measure Q, not under real-world probabilities; Q is constructed in Section 3
- **Variance**: $f(S) = -2\ln(S/F)$ has $f''(K) = 2/K^2$: the strip with weights $1/K^2$ is the variance swap and the VIX (Section 8)
  - Low strikes get the largest weights, so an expensive put wing makes variance expensive
- **Density**: a butterfly $[C(K-h) - 2C(K) + C(K+h)]/h^2 \to \partial^2 C/\partial K^2 = e^{-r\tau} q(K)$ Breeden & Litzenberger (1978)
  - h: the strike step; $q(K)$: the risk-neutral density of S_T; the density is a price (Section 7)
- Straddles, spreads and protective puts are finite strips with f'' concentrated at one or two strikes; moments and SVIX (simple VIX, the variance index of Martin (2017)): Appendix

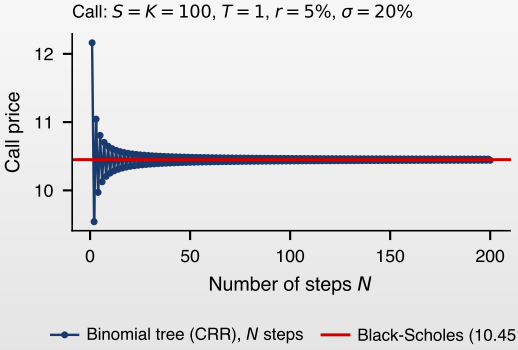
The Binomial Tree: How Do We Price an Option by Replication?

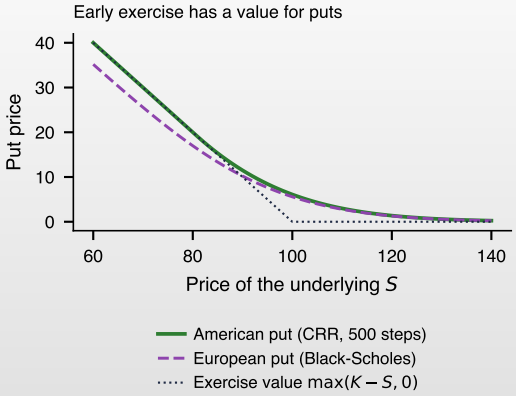
- One period of length Δt: $S \to uS$ or dS, with $d < R = e^{r\Delta t} < u$ (u, d: up and down factors; R: gross riskless return)
- Replicate the call with Δ shares and cash B: $\Delta uS + RB = C_u$ and $\Delta dS + RB = C_d$, where C_u, C_d are the call values after an up and a down move
  - Solution: $\Delta = \dfrac{C_u - C_d}{(u-d)S}$, and the price is $C = \Delta S + B = \dfrac{q\,C_u + (1-q)\,C_d}{R}$
  - $q = \dfrac{R-d}{u-d} \in (0,1)$ (here a probability, not the dividend yield) is the unique probability under which S_t/R^t is a martingale
- The real probability of an up move never enters: risk preferences are already in the price S
- **Fundamental theorem**: no arbitrage $\Leftrightarrow$ an equivalent martingale measure Q exists Harrison & Pliska (1981), Delbaen & Schachermayer (1994)
  - Q is unique exactly when the market is complete; with jumps or stochastic volatility there are many Q, and option prices carry risk premia (Section 8)

The CRR Tree: N Steps and Convergence to Black–Scholes

- Why N steps: one period has only two final states; with N steps the $N+1$ final states approximate the distribution of S_T
- The CRR (Cox–Ross–Rubinstein) tree Cox, Ross & Rubinstein (1979): $\Delta t = T/N$, $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$
  - This choice gives the one-step log return the variance $\sigma^2\Delta t$ (up to terms of order Δt^2), so the tree converges to the geometric Brownian motion of the Black–Scholes model
  - $q = (e^{r\Delta t} - d)/(u - d)$ is the same at every node
- European call: $C_0 = e^{-rT} \sum_{j=0}^{N} \binom{N}{j} q^j (1-q)^{N-j} \left(S_0 u^j d^{N-j} - K\right)^+$
  - j: the number of up moves; the sum is the discounted risk-neutral mean of the pay-off
- American options: value backwards from T, taking max(continuation value, exercise value) at every node

From the Tree to Black–Scholes





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Convergence and the Early-Exercise Premium

- Left: CRR price of a European call with $S_0 = K = 100$, $T = 1$, $r = 5\%$, $\sigma = 20\%$, for $N = 1, \ldots, 200$ steps; right: American vs European put
- Converges to Black–Scholes (10.451): $N = 50$: 10.411; $N = 200$: 10.441; $N = 1000$: 10.449
  - The error is of order $1/N$ and oscillates: $N = 10$ gives 10.253, $N = 11$ gives 10.611 (strike on or between nodes; Seminar 12, A6)
- American put at $S = K = 100$: 6.090 vs European 5.574; early-exercise premium 0.516: deep in the money it pays to receive K now

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The Black–Scholes Equation: What Price Does Continuous Replication Give? (1/2)

- Geometric Brownian motion $dS_t = \mu S_t\,dt + \sigma S_t\,dW_t$: log returns over τ follow the Normal distribution with variance $\sigma^2\tau$; constant σ and r, continuous trading, no costs, short selling allowed
  - $V(S,t)$: option value; V_t, V_S, V_{SS}: its partial derivatives in t and S (first, first, second)
  - Replicate $V(S,t)$ with a **self-financing** portfolio Π of Δ_t shares and cash B_t at r: $d\Pi = \Delta\,dS + rB\,dt$
- Itô's lemma: $dV = \left(V_t + \mu S V_S + \frac{1}{2}\sigma^2 S^2 V_{SS}\right)dt + \sigma S V_S\,dW_t$

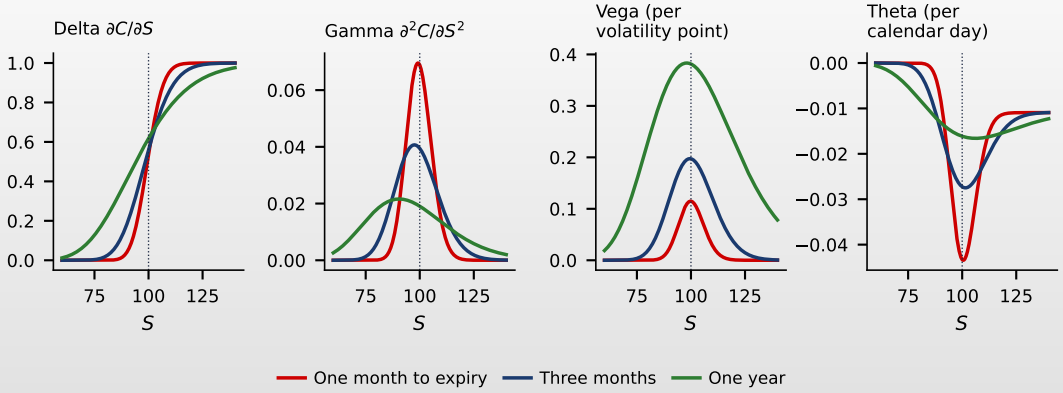
The Black–Scholes Equation: What Price Does Continuous Replication Give? (2/2)

- Choose $\Delta = V_S$, start from $\Pi = V$ ($B = V - SV_S$): $d(V - \Pi) = \left(V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} - rB\right) dt$; both dW_t and μ drop out
- A riskless position that costs nothing must earn nothing, hence the **Black–Scholes equation** Black & Scholes (1973), Merton (1973)
  - $V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} + rSV_S - rV = 0$, with $V(S, T) =$ pay-off
- That μ is irrelevant is a result, not an assumption; Feynman–Kac: $V = e^{-r\tau} E^Q[\text{pay-off}]$ with S drifting at r under Q

The Black–Scholes Formula

- European call and put, no dividends (with yield q: replace S by $Se^{-q\tau}$):
  - $C = S\,\Phi(d_1) - Ke^{-r\tau}\,\Phi(d_2), \qquad P = Ke^{-r\tau}\,\Phi(-d_2) - S\,\Phi(-d_1)$
  - $d_1 = \dfrac{\ln(S/K) + (r + \sigma^2/2)\tau}{\sigma\sqrt{\tau}}, \qquad d_2 = d_1 - \sigma\sqrt{\tau}$, with Φ the standard Normal distribution function
- Reading the two terms:
  - $\Phi(d_2)$ = risk-neutral probability that the call ends in the money
  - $\Phi(d_1)$ = number of shares in the replicating portfolio (the delta)
- Five inputs: S, K, τ, r observed; σ is the only one not observed: it must be estimated or implied

The Greeks: One Identity, All Sensitivities (1/2)



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The Greeks: One Identity, All Sensitivities (2/2)

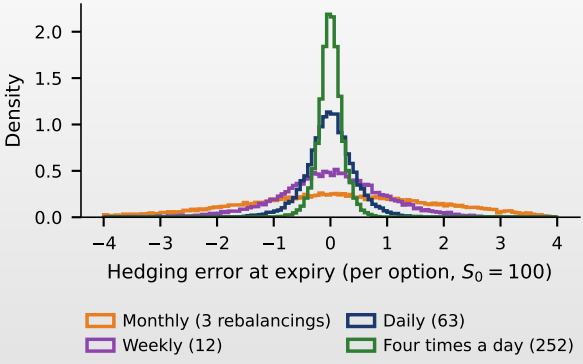
- Greeks: partial derivatives of the price, $\Delta = \partial C/\partial S$, $\Gamma = \partial^2 C/\partial S^2$, vega $\mathcal{V} = \partial C/\partial \sigma$, $\Theta = \partial C/\partial t$, rho $= \partial C/\partial r$
- Delta $\Delta = \Phi(d_1)$ (put: $\Phi(d_1) - 1$); gamma $\Gamma = \varphi(d_1)/(S\sigma\sqrt{\tau})$, φ the standard Normal density; key identity $S\varphi(d_1) = Ke^{-r\tau}\varphi(d_2)$
- Vega $= S\varphi(d_1)\sqrt{\tau} = \Gamma S^2\sigma\tau$: for one expiry gamma and vega are proportional (Seminar 12, A1, A7); theta $\Theta = rC - rS\Delta - \frac{1}{2}\sigma^2 S^2\Gamma$; rho $= K\tau e^{-r\tau}\Phi(d_2)$
- Call with $K = 100$; short maturities concentrate gamma and theta at the strike, long maturities carry the most vega; vega per vol point, theta per calendar day

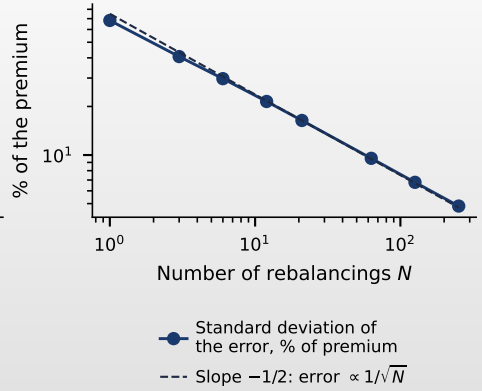
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Discrete Delta Hedging: How Much Risk Is Left with N Rebalancings?

- Black–Scholes replication needs continuous rebalancing; in practice the hedge is adjusted N times, at $t_i = iT/N$
- At $t_{\mathbf{0}} = 0$: sell the call at its Black–Scholes price $C_{\mathbf{0}}$, buy $\Delta_{\mathbf{0}} = \Phi(d_{\mathbf{1}})$ shares and keep the cash $B_{\mathbf{0}} = C_{\mathbf{0}} - \Delta_{\mathbf{0}} S_{\mathbf{0}}$
- At t_i, $i = 1, \ldots, N-1$: $\Delta_i = \Phi\big(d_{\mathbf{1}}(S_{t_i}, T - t_i)\big)$ and $B_i = B_{i-\mathbf{1}} e^{rT/N} - (\Delta_i - \Delta_{i-\mathbf{1}}) S_{t_i}$, so the portfolio is self-financing
- **Hedging error** at expiry: $\varepsilon_N = \Delta_{N-\mathbf{1}} S_T + B_{N-\mathbf{1}} e^{rT/N} - (S_T - K)^+$
- Simulation: geometric Brownian motion with $\mu = 8\%$ and $\sigma = 20\%$; $S_{\mathbf{0}} = K = 100$, $r = 4\%$, $T = 0.25$; 20 000 paths; $N = 3, 12, 63, 252$
- Why $\mu \neq r$: replication never uses μ; a real drift different from r shows that the error stays centred at 0

Discrete Delta Hedging: Algorithm and Error (1/2)





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Discrete Delta Hedging: Algorithm and Error (2/2)

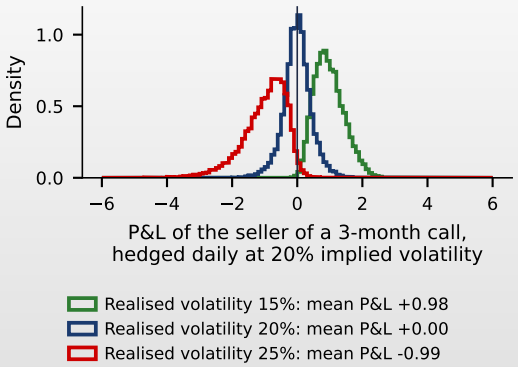
- Left: distribution of the error ε_N for four rebalancing frequencies; right: standard deviation of the error, in % of the premium, against N on log axes
- Black–Scholes premium 4.49 and $d_1 = 0.15$
- Mean error near 0; standard deviation (s.d.) in % of the premium: monthly 40.8%, weekly 21.5%, daily 9.5%, four times a day 4.8%; worst 1% of paths, in currency units per option: -2.68 (weekly), -1.17 (daily)
- Log-log slope -0.48; theory (Appendix): s.d. $\approx \sqrt{\pi/4}\,\mathcal{V}\sigma/\sqrt{N}$ ($\mathcal{V}$: vega), 1.01, 0.44, 0.22 vs simulated 0.96, 0.43, 0.22 ($N = 12, 63, 252$); costs grow like N

MFM_ch12_hedge_error

Derivation: the Gamma-Weighted Volatility P&L

- Sell the option at σ_{imp} (V^{imp}: its Black–Scholes value at σ_{imp}) and hold $\Delta = \partial_S V^{imp}$; the price moves with σ_{real}: $dS = \mu S\,dt + \sigma_{real} S\,dW_t$
- Itô on V^{imp}: $dV = V_t\,dt + \Delta\,dS + \frac{1}{2}\Gamma\sigma_{real}^2 S^2 dt$; the equation at σ_{imp} gives $V_t = rV - rS\Delta - \frac{1}{2}\sigma_{imp}^2 S^2\Gamma$
- Short option + Δ shares + cash at r, self-financing: $d\,\text{P\&L} = r\,\text{P\&L}\,dt + \frac{1}{2}\,\Gamma_t S_t^2\left(\sigma_{imp}^2 - \sigma_{real}^2\right)dt$, so over the life
  - $\text{P\&L} = \int_0^T e^{r(T-t)}\,\frac{1}{2}\,\Gamma_t S_t^2\left(\sigma_{imp}^2 - \sigma_{real,t}^2\right)dt$: path-dependent, weighted by Γ, large near K and near T
- A short variance swap pays $\sigma_{imp}^2 - \sigma_{real}^2$ with constant weights; a delta-hedged option is a variance bet with random weights
- First-order size: vega $\times$ (implied $-$ realised), checked on the next slide
  - Why: vega $= \Gamma S^2\sigma\tau$ (Section 4), so $\frac{1}{2}\Gamma S^2(\sigma_{imp}^2 - \sigma_{real}^2)\tau = \frac{1}{2}\Gamma S^2(\sigma_{imp} + \sigma_{real})(\sigma_{imp} - \sigma_{real})\tau \approx \Gamma S^2\sigma(\sigma_{imp} - \sigma_{real})\tau = \text{vega} \times (\sigma_{imp} - \sigma_{real})$

Selling Volatility at 20%: Three Outcomes



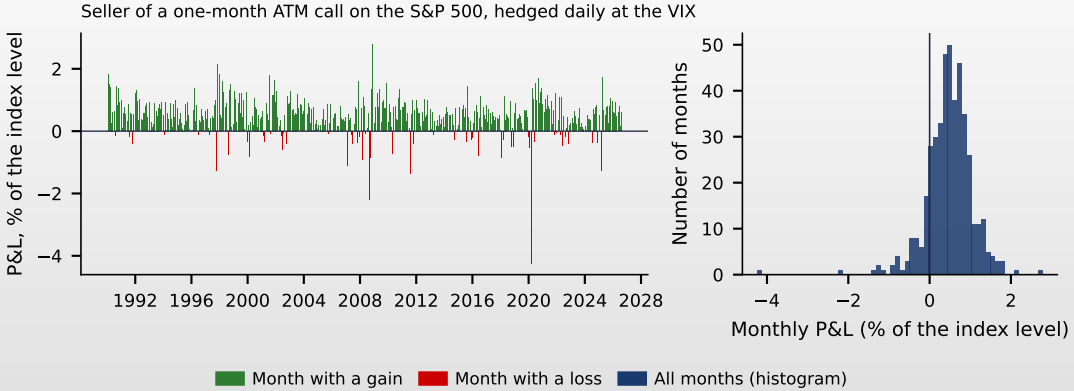
- Three-month call sold at 20% implied volatility
  - Hedged daily; realised volatility 15%, 20% or 25%
- Realised 15%: mean P&L $+0.98$, positive on 99.8% of paths
- Realised 20%: mean $+0.00$
  - Only discrete-hedging noise remains (s.d. 0.43)
- Realised 25%: mean -0.99, positive on only 2.2% of paths
- Each mean is close to vega $\times$ 5 vol points $= 0.197 \times 5 \approx \pm 0.99$
  - A clean bet on volatility

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Selling S&P 500 Volatility: the Strategy Tested

- The question: is implied volatility priced, on average, above realised volatility? A sold, delta-hedged option answers it directly
- Every 21 trading days t_0: sell a 21-day ATM call ($K = S_{t_0}$), priced by Black–Scholes with $\sigma = VIX_{t_0}/100$
- Hedge it daily with $\Delta_i = \Phi(d_1)$ at the same volatility; interest and dividends are ignored ($r = q = 0$)
- Monthly P&L, in % of the index: $\text{P\&L} = 100\left[C_0 + \sum_{i=0}^{20} \Delta_i \left(S_{t_{i+1}} - S_{t_i}\right) - (S_{t_{21}} - K)^+\right]/S_{t_0}$
- Why the VIX: it is the 30-day implied volatility available daily since 1990 in the course data
- Why ATM: vega is largest near the money, so the P&L mainly measures volatility

Thirty-Six Years of Selling S&P 500 Volatility (1/2)



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Thirty-Six Years of Selling S&P 500 Volatility (2/2)

- Left: P&L of every month, in % of the index (green: gain, red: loss); right: histogram of the monthly P&L
- 440 months from 1990-01: mean 0.48% a month, Newey–West $t = 16.0$ Newey & West (1987); positive in 86% of months; VIX above the realised volatility of the month in 84% of months (19.3 vs 15.3)
- Worst month -4.25% (2020-03), skewness -1.46: many small gains, a few large losses; consistent with a **volatility risk premium**; on real option quotes, delta-hedged option gains are negative for the buyer: the volatility risk premium is negative Bakshi & Kapadia (2003)

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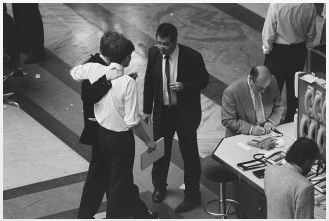
Implied Volatility: Which σ Explains the Market Price?

- **Implied volatility** σ_{imp}: the σ that makes the Black–Scholes price equal the market price
  - $BS(S, K, \tau, r, \sigma_{imp}) = C^{market}$; $BS(\cdot)$: the Black–Scholes formula, C^{market}: the observed price
  - The solution is unique because vega > 0: the price increases in σ
  - There is no closed form: σ enters both $\Phi(d_1)$ and $\Phi(d_2)$, so the equation is solved numerically
- Traders quote options in implied volatility, not in money: it is comparable across strikes, maturities and assets
- If Black–Scholes were right, all options on one asset would have the **same** implied volatility

Implied Volatility: Newton Inversion

- Newton's method: $\sigma_{k+1} = \sigma_k - \dfrac{BS(\sigma_k) - C^{market}}{\mathcal{V}(\sigma_k)}$
  - σ_k: the k-th iterate; $\mathcal{V}(\sigma_k)$: vega at σ_k, the slope of the price in σ
  - Why Newton: vega is the exact derivative of the price in σ, so convergence is quadratic near the solution
- Example: $S = K = 100$, three months, $r = 2\%$, market price 4.50; start at $\sigma_0 = 20.00\%$
  - First step: $BS(\sigma_0) = 4.2322$, $\mathcal{V} = 19.848 \Rightarrow \sigma_1 = 21.3495\%$
  - The second step changes σ by less than 10^{-6}: $\sigma_{imp} = 21.3495\%$
- When vega is close to zero (deep in or out of the money, very short expiries), a Newton step can leave the no-arbitrage range: then use a bracketing solver

After October 1987: the Smile Appears



Amsterdam stock exchange after the crash, 21 October 1987

Photo: Bart Molendijk / Anefo (1987); CC0; Wikimedia Commons

- Before 1987 index options had a nearly flat implied volatility across strikes Rubinstein (1994)
  - Since the crash, low-strike puts are much more expensive: the **skew**
- October 1986 – August 1987: option prices showed crash fears Bates (1991)
  - Not strongly in the two months immediately before the crash
- **Smile**: implied volatility as a function of K; **term structure**: as a function of T; **surface**: both

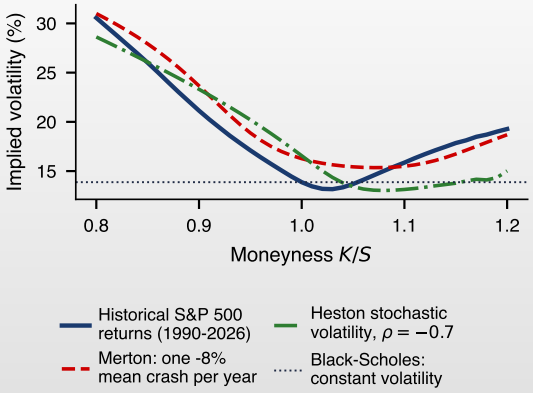
Why Smiles Exist and How Models Produce Them

- **Fat tails**: large moves are more frequent than the Normal distribution allows (Chapter 1), so deep OTM options are worth more than Black–Scholes says
- **Leverage and volatility feedback**: when prices fall, volatility rises Christie (1982); negative skewness makes low-strike puts dearer
- **Crash fear**: the market prices a negative jump risk, visible in put prices Bates (1991)
- **Demand for insurance**: investors buy puts, and intermediaries, who cannot fully hedge the risk, charge a premium for them Gârleanu, Pedersen & Poteshman (2009)

Model	Idea	Fits best
Jumps Merton (1976)	Poisson jumps added to the Brownian motion	steep short-dated smiles
Stochastic volatility Heston (1993), Hull & White (1987)	variance follows its own mean-reverting process, correlated with S	skew and term structure of longer maturities
Local volatility Gatheral (2006)	$\sigma(S, t)$ read off the whole surface (the formula of Dupire (1994))	matches every quoted price today, exotic options
Parametric smile, SVI (stochastic volatility inspired) Gatheral & Jacquier (2014)	five parameters per expiry, no dynamics	interpolation and quoting (Section 7)

- Processes: Chapter 11; the implied-volatility surface in detail: Franke, Härdle & Hafner (2019), Čížek, Härdle & Weron (2011)

Three Sources of a Smile (1/3)



MFM_ch12_smile_models

Three Sources of a Smile (2/3)

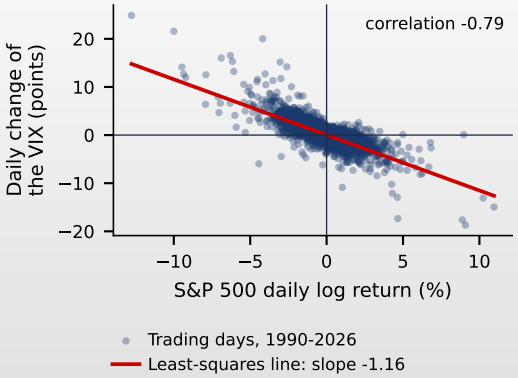
- One-month implied volatility from three sources
  - History: S&P 500 monthly returns rescaled to a martingale ($r = 0$, no risk adjustment)
  - A jump model and a stochastic-volatility model
- History: 9 225 overlapping returns, skewness -1.27, excess kurtosis 6.8
  - 30.6% at $K/S = 0.8$, 13.9% ATM, 19.3% at 1.2

Three Sources of a Smile (3/3)

- Merton jumps, one crash of about -8% a year
  - 31.0% at 0.8 vs 16.3% ATM
- Heston, $\rho = -0.7$
  - 28.6% at 0.8, 16.6% ATM, 15.0% at 1.2
- Risk-neutral laws: more left-skewed than realised Jackwerth (2000)
  - Priced crash risk steepens market smiles further

MFM_ch12_smile_models

The Leverage Effect in the Data



- Daily S&P 500 log return (%) vs the change of the VIX (index points) on the same day, 1990–2026
- Correlation -0.79 over 9 244 days: implied volatility rises when the market falls
- Slope -1.16 VIX points per 1% return
  - Steeper on down days (-1.39) than on up days (-1.01)
  - Mean VIX change: $+0.88$ on down days, -0.76 on up days
- The $\rho < 0$ of Heston and the asymmetry of GJR-GARCH, the Glosten–Jagannathan–Runkle version of the GARCH conditional-variance model (Chapter 5)
  - Low-strike puts gain from the fall and from the rise in volatility
  - Hence they are expensive

MFM_ch12_leverage

The Bitcoin Surface: How to Describe the Smile Without Arbitrage?

- Deribit lists European options on Bitcoin, traded 24/7 Deribit DVOL
  - The options are **inverse**: premium and pay-off in bitcoin (BTC), not in US dollars (USD)
  - Each expiry has its own forward F
  - DVOL: Deribit's 30-day implied-volatility index
- Our snapshot: 22 September 2026, 07:51 UTC, with 9 expiries that have at least two days left and 375 OTM options with a positive bid
- Why Bitcoin: it is the only asset in the course data with a full option chain across strikes and expiries
- Hedging Bitcoin options: Matic, Packham & Härdle (2023)

Inverse Options: Why the USD Formulas Still Apply

- An inverse call pays $(S_T - K)^+/S_T$ BTC at T; converted at S_T USD per BTC this is exactly $(S_T - K)^+$ USD, the pay-off of a standard USD call
  - Toy example: $K = 90\,000$, $S_T = 100\,000$: the holder receives $10\,000/100\,000 = 0.1$ BTC, worth 10 000 USD
- Same pay-off, same value: a premium of p BTC paid today costs $p \times S$ USD at today's index S, so $p \times S$ is the USD price of the standard call
  - Black–Scholes/Black-76 and the exchange's mark IV therefore apply to $p \times S$ unchanged; this is the conversion used in the smile, density and VIX slides
- The rate: Deribit quotes a forward F for each expiry; this chapter uses $e^{r\tau} = F/S$, the carry implied by the futures basis, in place of a Treasury rate
  - Student check: an inverse put with $K = 90\,000$ ends at $S_T = 75\,000$. What does it pay in BTC and in USD? (Answer: 0.2 BTC, i.e. 15 000 USD)

The SVI Smile: Total Variance and Parameters

- Work with the **total implied variance** $w(k) = \sigma_{imp}^2(k)\,\tau$ at log-moneyness $k = \ln(K/F)$
- SVI parameterisation (stochastic volatility inspired) Gatheral & Jacquier (2014):
  $w(k) = a + b\left(\rho\,(k-m) + \sqrt{(k-m)^2 + s^2}\right)$
  - a: level
  - b: slope of the wings
  - ρ: tilt (negative = put skew)
  - m: shift
  - s: curvature at the bottom

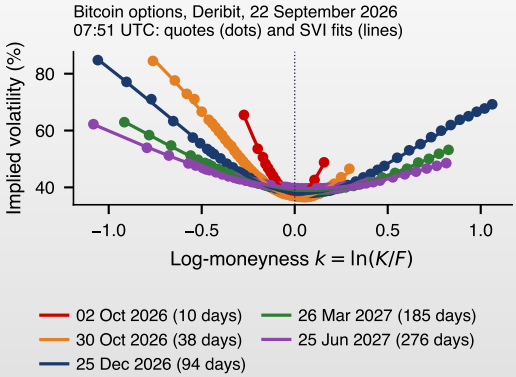
The SVI Smile: Why This Shape and How We Fit It

- Why w and k: the no-arbitrage conditions (next slide) are written directly in $w(k)$, and different expiries become comparable
- Why SVI: as $|k| \to \infty$, w becomes linear in k, the shape that Lee (2004) allows for the wings; five parameters are estimated from 24–58 quotes per expiry
- Wing slopes: $b(1-\rho)$ on the left and $b(1+\rho)$ on the right
- Fit: for each expiry, $\min \sum_j \left(w(k_j) - \sigma_{imp,j}^2 \tau\right)^2$ (least squares on w), with parameter bounds and six starting points
- Each fit is then checked against the no-arbitrage constraints

No-Arbitrage Constraints on the Smile

- **Wings**: Lee (2004) bounds the asymptotic slope, $\limsup_{|k|\to\infty} w(k)/|k| \le 2$; for SVI $b(1+|\rho|) \le 2$ is necessary, and $w \ge 0$ needs $a + bs\sqrt{1-\rho^2} \ge 0$
  - A steeper wing means fatter tails of S_T under Q (moment formula: Appendix)
- **Butterfly** (call prices convex in K, density $q \ge 0$), with $w' = \partial_k w$, $w'' = \partial_k^2 w$ Gatheral & Jacquier (2014): $g(k) \ge 0$ is exactly a nonnegative density
  - $g(k) = \left(1 - \frac{kw'}{2w}\right)^2 - \frac{w'^2}{4}\left(\frac{1}{w} + \frac{1}{4}\right) + \frac{w''}{2} \ge 0$ for all k, plus $d_+(k) = -k/\sqrt{w} + \sqrt{w}/2 \to -\infty$ as $k \to \infty$ (Gatheral & Jacquier (2014), Lemma 2.2)
- **Calendar**: $\partial_\tau w(k,\tau) \ge 0$ for every k: total variance cannot fall with maturity
- Slice-by-slice least squares imposes none of them; SSVI, the surface version of SVI, has simple sufficient conditions (Gatheral & Jacquier (2014), Theorems 4.1–4.2)

The Bitcoin Smile



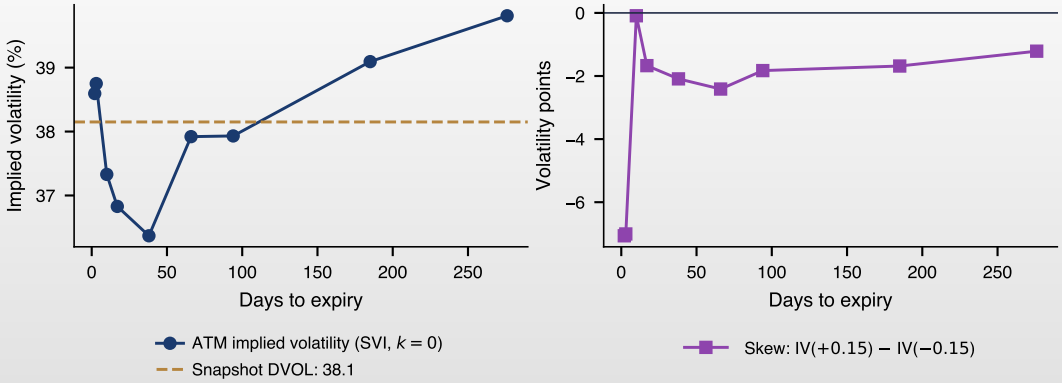
- Dots: mark implied volatilities of OTM options; lines: SVI fits, five expiries
  - Mark IV (implied volatility): the reference volatility computed by the exchange
- A true **smile**: both wings rise
  - Unlike the one-sided equity skew
  - Bitcoin can jump up as well as down Zulfiqar & Gulzar (2021)
- Short expiries have the steepest wings
  - A jump over a few days weighs more relative to $\sigma\sqrt{\tau}$
- SVI root mean squared error: at most 0.70 vol points
  - Puts dearer than calls at the same $|k|$: the skew is negative at all 9 expiries

MFM_ch12_btc_smile

Are the Nine Bitcoin Fits Arbitrage-Free?

- For 2 October, 9 October and 27 November, the parameter a hits the lower bound imposed in the optimisation (-1): the individual parameters are not precisely identified
- Wings: largest slopes 0.51 (right) and 2.30 (left), the left wing exceeds Lee's upper bound of 2
- Butterfly: $g(k) < 0$ somewhere on the grid $k \in [-1.5, 1.5]$ for 2 of 9 expiries (lowest -0.017, 2 October 2026)
- Calendar: 1 of 8 adjacent pairs crosses: $w(\text{2 October 2026}) > w(\text{9 October 2026})$ on 9% of $k \in [-0.5, 0.5]$, by at most 0.0018
  - Separate unconstrained fits can violate cross-expiry restrictions; these fitted curves do not establish arbitrage in executable market quotes
- Remedy: estimate the surface jointly under the constraints (SSVI, or least squares with $g \geq 0$ and $\partial_\tau w \geq 0$, also imposing the wing conditions)

Term Structure and Skew



MFM_ch12_btc_term

Expiry, Skew and the DVOL Index

- Left: ATM implied volatility by expiry and the DVOL index; right: skew $= \mathrm{IV}(k = +0.15) - \mathrm{IV}(k = -0.15)$, with IV the implied volatility at that log-moneyness
- ATM between 36.4% and 39.8%; longest expiry (276 days) 39.8%: the curve falls at short maturities, then rises beyond about one month
- DVOL 38.1 vs ATM 36.8% and 36.4% around 30 days: a variance-strip index need not lie between ATM volatilities
- Skew negative at all 9 expiries: closest to 0 at 10 days (-0.1), most negative at 66 days (-2.4), then -1.2 at 276 days
  - At 2 and 3 days, $k = \pm 0.15$ lies beyond the quotes (at most $k = 0.11$ and 0.14): the skew of about -7 points is an SVI extrapolation there
- One snapshot is one day

MFM_ch12_btc_term

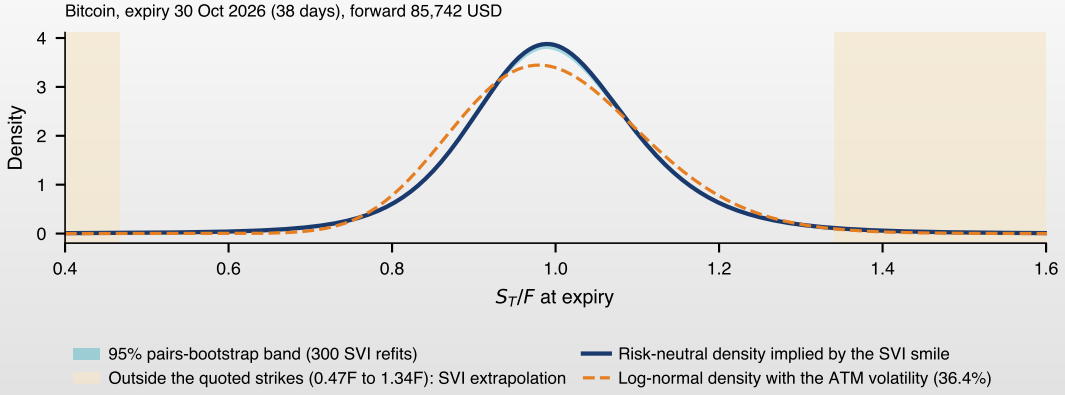
The Risk-Neutral Density: From Prices to a Density (1/2)

- Recall Breeden & Litzenberger (1978) (Section 2): the risk-neutral density q of S_T is a second derivative of call prices
  - $q(K) = e^{r\tau}\,\partial^2 C(K)/\partial K^2$; $q \geq 0$ is the butterfly condition of Section 7
- From noisy, discrete quotes this is an ill-posed inverse problem: differentiating twice amplifies the noise; fit a smooth smile (SVI), then differentiate (other estimators: Appendix)
- Uncertainty: pairs bootstrap of the quotes, refit, differentiate: pointwise bands for q and intervals for tail probabilities

The Risk-Neutral Density: From Prices to a Density (2/2)

- Beyond the last quoted strike the data identify nothing: the tail is an extrapolation; report where the quotes end
- q mixes probabilities and risk aversion
  - Pricing kernel $\propto q/p$, with p the physical density Aït-Sahalia & Lo (1998), Grith, Härdle & Park (2013)
  - Its non-monotone shape is the pricing-kernel puzzle Jackwerth (2000)

What the Market Prices for Bitcoin (1/3)



MFM_ch12_btc_rnd

What the Market Prices for Bitcoin (2/3)

- Density from the SVI smile of the expiry closest to 30 days
  - 38 days, forward 85 742 USD, ATM 36.4%; 95% pairs-bootstrap band; log-normal benchmark
  - Shaded: outside the 56 quotes ($0.47F$ to $1.34F$)
- Fall of more than 30%: 1.21% $[1.08, 1.24]$ vs 0.14% log-normal, about 8 times more
  - Fall of more than 20%: 4.28% vs 3.27%; rise above 20%: 5.3% vs 5.3%

What the Market Prices for Bitcoin (3/3)

- Narrow band: $\pm 2.0\%$ at the centre, $\pm 7\%$ at $0.7F$
  - Smoothed mark volatilities: fitting noise, not bid–ask uncertainty
  - 0.13 of the 1.21 points lie below the lowest quoted strike
- Prices of insurance, not forecasts: they include the premium for crash risk

MFM_ch12_btc_rnd

Variance Swaps: What Does Future Variance Cost?

- A **variance swap** pays at T: notional $\times$ (realised variance $-$ strike K_{var})
  - Realised variance $\frac{252}{n}\sum r_t^2$, r_t: the n daily log returns to T
  - Fair strike: the risk-neutral expected variance Demeterfi, Derman, Kamal & Zou (1999)
  - $K_{var} = \frac{2e^{r\tau}}{\tau}\left(\int_0^F \frac{P(K)}{K^2}\,dK + \int_F^\infty \frac{C(K)}{K^2}\,dK\right)$ (spanning with $f = -2\ln(S/F)$, Section 2)
- Why it works: Itô on $\ln S$ (continuous paths, $r = q = 0$): $-2\ln(S_T/F) = \int_0^T \sigma_t^2\,dt - 2\int_0^T dS_t/S_t$, and the last term is a Q-martingale Britten-Jones & Neuberger (2000)
  - "Model-free" means free of a model for σ_t, not of continuity: with crash jumps the strip understates expected realised variance Carr & Wu (2009) (Appendix)
- Buyer of variance = insurance against turbulence; seller = earns a premium, pays in crashes Carr & Wu (2009)

The VIX Index: the Cboe Formula

- The discrete version of the strip K_{var} from the previous slide Cboe (2025): $\sigma^2 = \frac{2}{T}\sum_i \frac{\Delta K_i}{K_i^2} e^{RT} Q(K_i) - \frac{1}{T}\left(\frac{F}{K_0} - 1\right)^2$
  - T: time to expiry in years
  - R: risk-free rate
  - K_i: the strikes
  - K_0: the first strike at or below the forward index level F
  - $Q(K_i)$: bid–ask mid of the put for $K_i < K_0$, of the call for $K_i > K_0$, the average of both at K_0
  - The second term corrects for the strip being split at K_0 rather than exactly at F

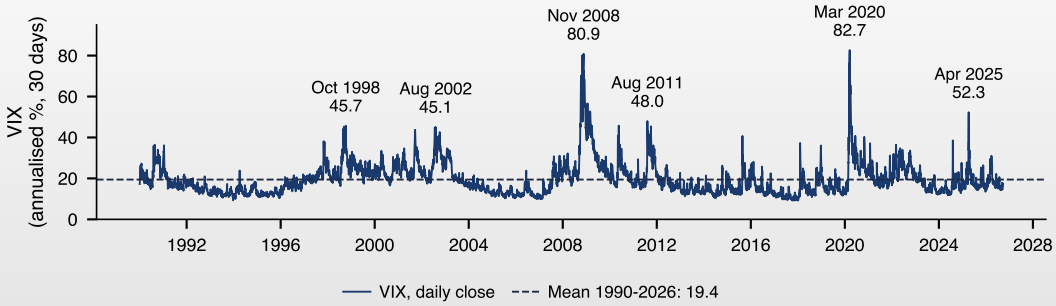
The VIX Index: Computation Rules

- Cboe Volatility Index (VIX), since 1993; since 2003 computed from S&P 500 options with the formula of the previous slide Cboe (2025)
- $\Delta K_i = (K_{i+1} - K_{i-1})/2$, one-sided at the two ends: at the lowest (highest) strike, ΔK is the full gap to the next higher (lower) strike
- Options with a zero bid are dropped, and each wing stops after two consecutive zero bids
- Two expiries around 30 days, interpolated to exactly 30 days; VIX $= 100\,\sigma$
- VIX futures since 2004, VIX options since 2006 Cboe (2025)
- The VIX is called the "fear gauge" Whaley (2000)

Worked Example: a VIX for Bitcoin

- Cboe rules on our Deribit snapshot: USD mid Q = BTC mid $\times$ index S; F and $e^{RT} = F/S$ from Deribit's forward (Cboe: F from parity, R from Treasury yields)
  - Near expiry: 17 days, 26 OTM options, $\sigma = 39.5\%$; next expiry: 38 days, 56 OTM options, $\sigma = 40.2\%$
- Interpolate the total variances $\sigma_i^2 T_i$ linearly in time to 30 days:
  - $\sigma_{30}^2 = \frac{1}{30}\left[T_1\sigma_1^2 \frac{T_2 - 30}{T_2 - T_1} + T_2\sigma_2^2 \frac{30 - T_1}{T_2 - T_1}\right]$ (days)
- Our Bitcoin VIX: 40.0; Deribit's DVOL at the same moment: 38.1 Alexander & Imeraj (2020); both exceed the ATM volatilities: the variance swap also pays for the wings
- The gap to DVOL, 1.9 vol points, cannot be attributed to truncation: the SVI extrapolation it would require violates Lee's wing condition (Appendix; Seminar 12, A8)

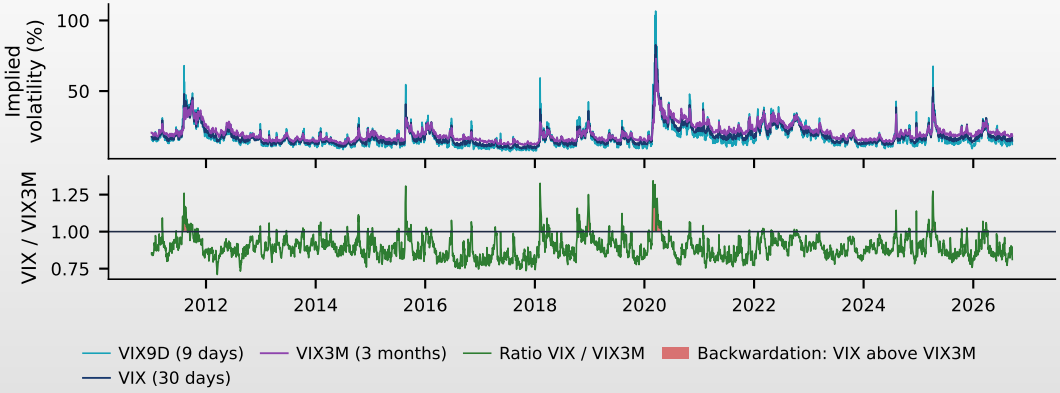
The VIX, 1990–2026



- Daily close with the six highest peaks at least 500 days apart
- Mean 19.4 (5.6% a month), range 9.1–82.7 (2020-03-16, COVID-19); above 30 on 7.9% of days: sudden spikes, slow decay, as in GARCH (Chapter 5)

MFM_ch12_vix_history

The VIX Term Structure (1/3)



MFM_ch12_vix_term

The VIX Term Structure (2/3)

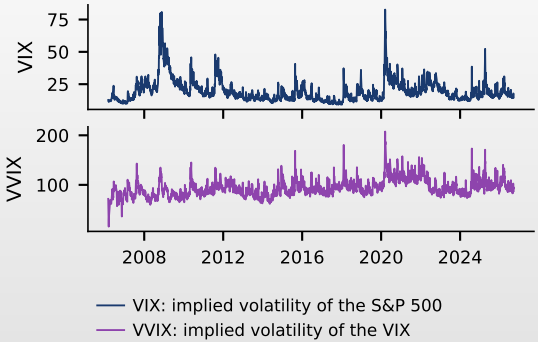
- Top: implied volatility, 9 days, 30 days, 3 months
  - Bottom: ratio VIX / VIX3M, 2011–2026
- Normally upward sloping (**contango**): mean VIX / VIX3M = 0.89; VIX9D below the VIX on 73% of days
- Inverted (**backwardation**) on 7.6% of days
  - Always in crises: fear is concentrated in the near term

The VIX Term Structure (3/3)

- Next 21 days: after backwardation vs in contango
  - S&P 500 log return over the next 21 days 2.88% vs 0.80%; realised volatility 26.7% vs 13.6%
  - High risk and high expected return after panics Bollerslev, Tauchen & Zhou (2009)
  - The comparison is descriptive: the 21-day windows overlap, and backwardation days are few and clustered in a handful of crises

MFM_ch12_vix_term

The Volatility of Volatility



- VIX (top) and VVIX (bottom), daily closes, 2006–2026
- Mean VVIX 93.5
  - In relative terms, the VIX is expected to move about 5.3 times more than the index
  - 5.3: the average daily ratio VVIX/VIX; both are annualised implied volatilities in %, VVIX of percentage changes of the VIX, VIX of percentage changes of the S&P 500
- Peak 207.6 on 2020-03-16, the day of the VIX peak
  - Daily changes of VIX and VVIX: correlation 0.71
- Volatility is itself random
  - The core of stochastic-volatility models such as Heston Heston (1993)

MFM_ch12_vvix

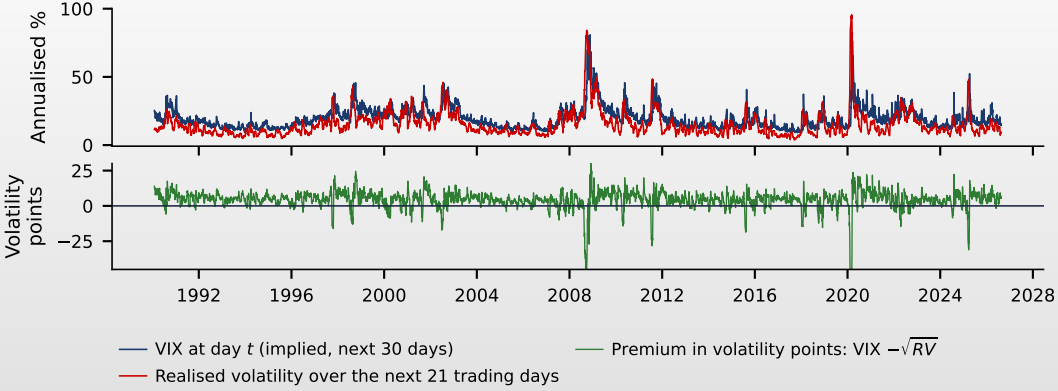
The Variance Risk Premium (1/2)

- Variance risk premium (VRP) Carr & Wu (2009), Bollerslev, Tauchen & Zhou (2009); three versions that must not be confused:
  - **Economic (ex ante)**: $E_t^Q[RV_{t,t+\mathbf{21}}] - E_t^P[RV_{t,t+\mathbf{21}}]$; $VIX_t^{\mathbf{2}}$ estimates the first term, the second needs a forecast
  - **Ex post** (used here): $VIX_t^{\mathbf{2}} - RV_{t,t+\mathbf{21}}$, the realised P&L of the seller of a variance swap, known only at $t+21$
  - **Ex-ante proxy**: $VIX_t^{\mathbf{2}} - RV_{t-\mathbf{21},t}$, past variance as the forecast of E_t^P; known at t, so it can serve as a predictor (Section 11)
  - $RV_{t,t+\mathbf{21}} = \frac{\mathbf{252}}{\mathbf{21}} \sum_{i=\mathbf{1}}^{\mathbf{21}} r_{t+i}^{\mathbf{2}}$ (annualised, in $\%^{\mathbf{2}}$), r_{t+i} the daily log returns in %
  - The VIX covers 30 calendar days, $RV_{t,t+\mathbf{21}}$ the next 21 trading days: roughly the same month, an approximation (holidays and the 365- vs 252-day annualisation make the windows differ slightly)

The Variance Risk Premium (2/2)

- Positive on average if investors dislike states with high volatility, which are also bad states for wealth
- Overlapping 21-day windows are autocorrelated: Newey–West with at least 21 lags Newey & West (1987); persistent volatility can extend the dependence, so check longer bandwidths

Implied vs Realised Volatility, S&P 500 (1/3)



MFM_ch12_vrp

Implied vs Realised Volatility, S&P 500 (2/3)

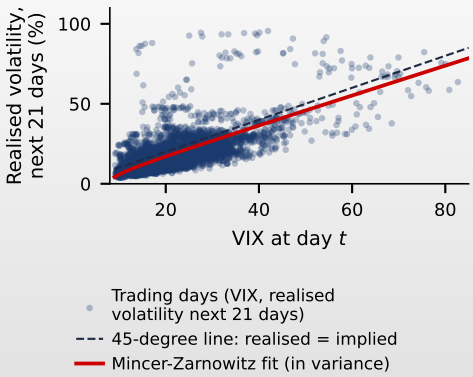
- Top: VIX and the volatility realised over the next 21 trading days
  - Bottom: their difference, 1990–2026, labelled VIX − RV: there RV is the realised volatility $\sqrt{RV_{t,t+\mathbf{21}}}$, in vol points, not the variance
- 9 203 days: mean VIX 19.4, mean realised 15.3
  - Premium 4.09 vol points (Newey–West standard error, SE, 0.27)
  - In variance: 113 $\%^{\mathbf{2}}$, $t = 5.5$

Implied vs Realised Volatility, S&P 500 (3/3)

- Positive on 86% of days; the exceptions are crashes
  - Worst -72.0 vol points, window from 2020-02-20 (COVID-19)
- Same message as the delta-hedged option
  - Sellers of volatility are paid on average and lose in crises

MFM_ch12_vrp

Is the VIX a Good Forecast?



- Mincer–Zarnowitz Mincer & Zarnowitz (1969): $RV_{t,t+\mathbf{21}} = \alpha + \beta\, VIX_t^{\mathbf{2}} + \varepsilon_t$
  - Unbiased: $\alpha = 0$, $\beta = 1$; Newey–West errors, 21 lags
- $\hat{\alpha} = -51.9$ (34.4), $\hat{\beta} = 0.86$ (0.10), $R^{\mathbf{2}} = 0.37$
  - Separately neither differs from $(0, 1)$: $t = -1.51$ and -1.34
- But the estimates correlate -0.90
  - Joint Wald $\chi^{\mathbf{2}}_{\mathbf{2}} = 41.7$, $p = 8.9 \cdot 10^{-\mathbf{10}}$: unbiasedness rejected
  - Chart axes in volatility (%); the fit is estimated in variance and drawn back in volatility; most points lie below the 45-degree line
- Past realised variance alone: $R^{\mathbf{2}} = 0.29$
  - With both: VIX keeps its weight (0.77), past variance adds little (0.08, SE 0.13)

MFM_ch12_vix_rv_mz

Forecast-Evaluation Tools in Plain Words

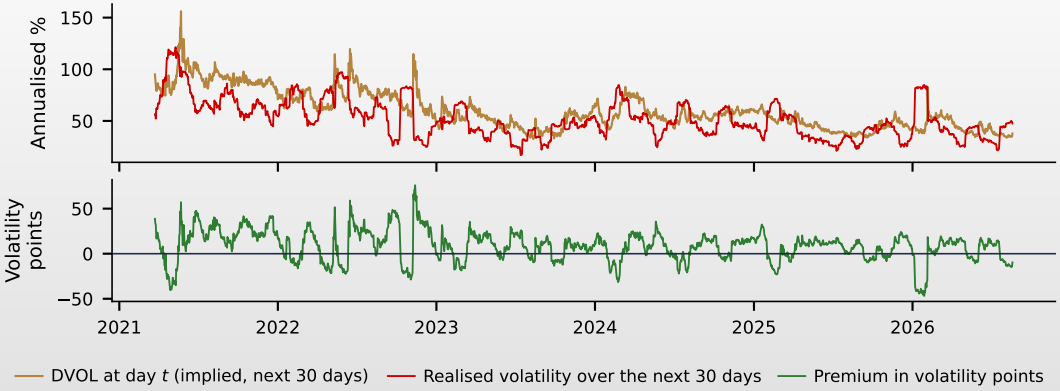
- In logs the VIX is close to an unbiased forecast, and out of sample it is not significantly better than a model built from past realised variance
- **Instrumental variables**: the lagged VIX^2 instruments the VIX^2 measured with error; a large first-stage t (regression on the instrument) means a strong instrument
- **HAR-RV**: forecast the realised variance of the next month by a regression on its averages over the past day, week and month
- **QLIKE** loss $L = x - \ln x - 1$ with $x = RV/\hat{\sigma}^2$: zero for a perfect forecast, and it punishes forecasts that are too low more than forecasts that are too high
  - Toy check: $RV = 100$; forecast 50 gives $x = 2$, $L = 0.31$; forecast 200 gives $x = 0.5$, $L = 0.19$
- **Diebold–Mariano**: a t-test that the mean loss difference between two forecasts is zero, with a HAC standard error because overlapping targets make the differences autocorrelated

Logs, Instruments and HAR-RV: Inference Done Right

Specification	$\hat{\beta}$	SE	$t(\beta = 1)$	Note
Levels, OLS	0.86	0.10	-1.34	joint Wald $p = 8.9 \cdot 10^{-10}$
Logs, OLS	1.06	0.035	1.65	intercept -0.94, $R^2 = 0.57$
Logs, instrumental variables	1.02	0.042	0.60	first-stage $t = 37.7$

- Levels are dominated by 2008 and 2020; in logs the slope is close to 1 and the premium is a proportional shift (intercept < 0)
- VIX^2 measures expected variance with error, which attenuates the ordinary least squares (OLS) slope
  - Instrument: VIX^2 21 days earlier Christensen & Prabhala (1998); the slope barely moves
  - Valid only if the lag is uncorrelated with the forecast error (untestable with one instrument)
- Against the heterogeneous autoregressive model HAR-RV Corsi (2009)
  - In sample the HAR terms add information: $\chi^2_3 = 14.2$, $p = 0.003$
  - Out of sample from 2000: VIX about 6% better in QLIKE (quasi-likelihood) loss, Diebold–Mariano $p = 0.35$ Diebold & Mariano (1995): not significant
  - Test details: ▸ Appendix

The Variance Risk Premium of Bitcoin (1/3)



MFM_ch12_vrp_btc

The Variance Risk Premium of Bitcoin (2/3)

- Top: DVOL and Bitcoin volatility realised over the next 30 calendar days (Bitcoin trades every day)
  - Bottom: their difference, 2021–2026
- 1 975 days: mean DVOL 60.3 vs realised 51.7
  - Premium 8.5 vol points (SE 1.6, $t = 5.3$)
  - About 2.1 times the S&P 500 premium in vol points, but smaller relative to the level: 14% of mean DVOL vs 21% of mean VIX (implied/realised 1.16 vs 1.27; different sample periods)

The Variance Risk Premium of Bitcoin (3/3)

- Positive on 71% of days, less often than the S&P 500
  - Worst -46.9 vol points (window from 2026-01-28)
- Short sample: 66 non-overlapping 30-day periods
  - A Bitcoin VIX and its premium: Alexander & Imeraj (2020)

MFM_ch12_vrp_btc

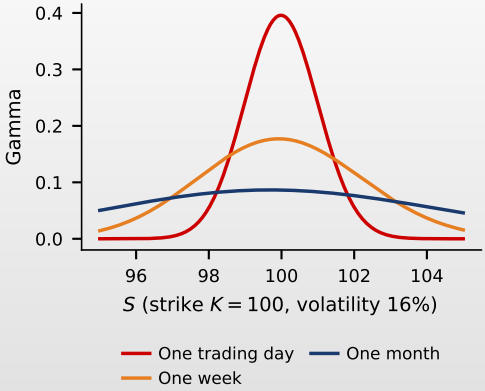
0DTE Options: What Changes When Expiry Is Today? (1/2)

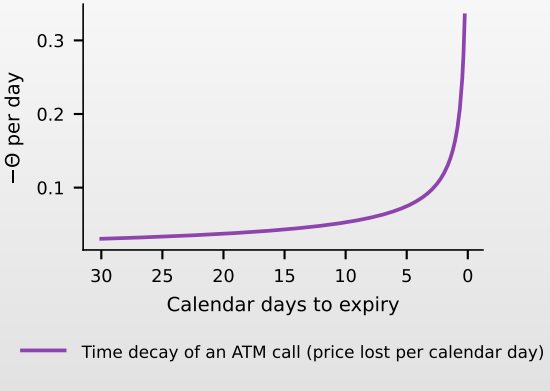
- **0DTE** (zero days to expiry) options: traded on the day they expire Cboe 0DTE
  - S&P 500 options now expire every trading day, so a 0DTE contract exists every day
- More than 75% of retail trades in S&P 500 options involve 0DTE contracts, and retail investors lose money on them Beckmeyer, Branger & Gayda (2023)
  - 0DTE options were 48% of index-option trading in August 2023 Brogaard, Han & Won (2023)

0DTE Options: What Changes When Expiry Is Today? (2/2)

- More 0DTE trading is followed by higher index volatility Brogaard, Han & Won (2023)
  - Mostly through speculative retail trades
  - Ultra-short pricing needs skewness and kurtosis corrections Bandi, Fusari & Renò (2023)
- Theory from Section 4: with $\tau \to 0$ gamma and theta explode at the money

Gamma and Time Decay Near Expiry (1/2)





MFM_ch12_0dte_gamma

Gamma and Time Decay Near Expiry (2/2)

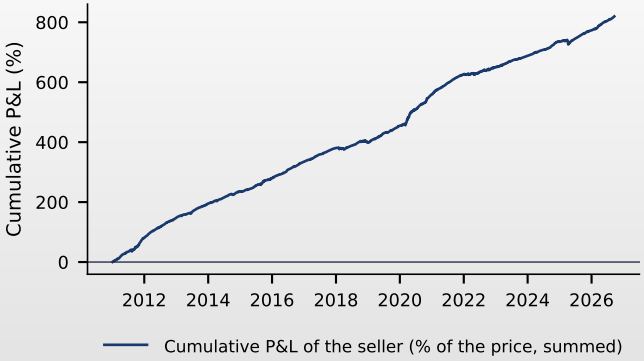
- Left: gamma of a call with $K = 100$ one day, one week and one month before expiry; right: daily time decay of an ATM call
- Inputs: $S = K = 100$, $\sigma = 16\%$, $r = 0$; gamma counts trading days ($\tau = 1/252$ and $21/252$), time decay counts calendar days ($\tau = 1/365$ and $30/365$), as on the two axes
- ATM gamma one day before expiry 0.396 vs 0.086 one month before: 4.6 times larger ($= \sqrt{21}$); a 1% move in the last day changes delta by about 0.4
- Time decay 0.167 in the last day vs 0.030 with 30 days left, about 5.5 times more: the seller collects fast decay and carries large gamma risk
  - The ATM value, about $0.4\, S\sigma\sqrt{\tau} = 0.33$, is lost entirely during that day, roughly twice the starting rate

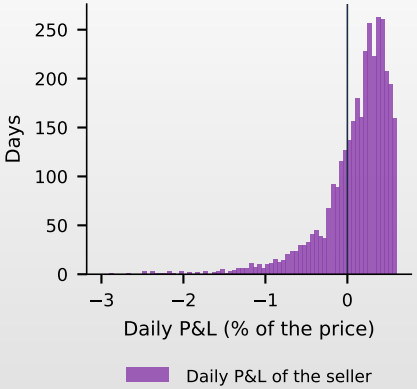
MFM_ch12_0dte_gamma

Selling a Same-Day Straddle: the Method

- The question: does the seller of an option expiring today also earn the volatility premium of one-month options?
- The course data contain no 0DTE prices, so we build a stylised test on SPY, 2011–2026 (VIX9D starts in 2011): sell an ATM straddle (a call and a put with $K = S_t^{open}$) at the open and settle it at the close
- Premium: $\Pi_t = \sqrt{2/\pi}\,\sigma_{day}\,S_t^{open}$, the approximate Black–Scholes value of an ATM straddle ($E|Z| = \sqrt{2/\pi}$ for a standard normal Z)
- $\sigma_{day} = \dfrac{VIX9D_{t-\mathbf{1}}}{100}\sqrt{\omega/252}$: the previous day's 9-day implied volatility, scaled to one session
- $\omega = 0.74$: the share of the session (open to close) in daily variance, estimated on 1993–2010, before the test period
- The seller's daily P&L, in % of the opening price: $100\big(\Pi_t/S_t^{open} - |S_t^{close}/S_t^{open} - 1|\big)$

Selling a Same-Day Straddle: a Stylised Test (1/2)





MFM_ch12_0dte_straddle

Selling a Same-Day Straddle: a Stylised Test (2/2)

- Left: the seller's cumulative P&L (sum of daily P&L, in % of the price); right: histogram of the daily P&L
- 3 950 days: mean premium 0.76% vs mean absolute move 0.55%; the seller wins on 75% of days, 20.7 basis points a day
- Worst day −8.25% (2025-04-09), about 11 times the mean premium; stylised pricing measures the premium inside VIX9D, not traded 0DTE prices Bandi, Fusari & Renò (2023), Beckmeyer, Branger & Gayda (2023)

MFM_ch12_0dte_straddle

Case Study: the Research Question (1/2)

- **Questions** Back, Crotty & Kazempour (2022): option prices give lower bounds on the equity premium
  - Are these bounds valid conditionally?
  - Are they tight conditionally?
  - Do they forecast returns out of sample?
- Why: a tight bound is a real-time equity-premium forecast with no estimated parameter; in the regression of realised returns on the bound, Martin (2017) could not reject $\alpha = 0$, $\beta = 1$ on 1996–2012, a test with low power

Case Study: the Research Question (2/2)

- **Martin bound** (eq. 3), $R_{t,T}$ the gross market return, $R_{f,t,T}$ the gross riskless return, var_t^* the risk-neutral variance ($\mathrm{SVIX}^2_{t,T}$ not annualised here):
  - $E_t[R_{t,T}] - R_{f,t,T} \geq R_{f,t,T}\,\mathrm{SVIX}^2_{t,T}, \qquad \mathrm{SVIX}^2_{t,T} = \mathrm{var}_t^*(R_{t,T}/R_{f,t,T})$
  - Valid if $\mathrm{cov}_t(M_{t,T}R_{t,T}, R_{t,T}) \leq 0$ ($M_{t,T}$ the stochastic discount factor); tight if this covariance is zero, e.g. a log-utility investor who holds the market
- **Chabi-Yo–Loudis bound** Chabi-Yo & Loudis (2020): built from the first four risk-neutral moments, free of preference parameters in its restricted form; almost always above the Martin bound (Figure 3.1)
- **Link to this chapter**: SVIX is the static-spanning strip with $f = S^2$ (Section 2 and ▸ Appendix)
  - Realised slackness $s_{t,T} = R^e_{t,T} - b_{t,T}$ (excess return minus bound): valid means $E_t[s_{t,T}] \geq 0$, tight means $E_t[s_{t,T}] = 0$

The Case-Study Tests in Plain Words

- **Kodde–Palm** D_1: the Σ^{-1}-weighted squared distance from the vector of sample moments to the region where all moments are ≥ 0; $D_1 = 0$ if all are already positive. D_2: how far the moments are from all being zero, beyond D_1; a large D_2 rejects tightness
  - Toy check with two moments and $\Sigma = I$: $\bar{\lambda} = (0.5, -0.2)$ gives $D_1 = 0.2^2 = 0.04$ and $D_2 = 0.29 - 0.04 = 0.25$; with $\bar{\lambda} = (0.5, 0.2)$, $D_1 = 0$ and $D_2 = 0.29$
- **Mixture of χ^2 laws**: under H_0 some inequality constraints bind and others do not, so the null law weights $\chi^2_0, \ldots, \chi^2_{10}$; Kodde & Palm (1986) tabulate lower and upper bounds for its critical values
- **Hansen–Hodrick covariance**: a HAC covariance that keeps, with equal weights, exactly the autocovariances created by the overlap of the return windows
- **Amihud–Hurvich correction**: a persistent predictor biases the OLS slope of a predictive regression (Stambaugh bias); the correction adds as a regressor the innovation of the predictor, estimated from a bias-corrected AR(1), reducing the bias

Case Study: Data and Method (1/2)

- **Data** (Section 3, Appendix C of the paper; filters: Appendix)
  - Daily Martin and Chabi-Yo–Loudis bounds, January 1990 – December 2020, from S&P 500 index options, interpolated to 30, 90, 180 and 360 days
  - S&P 500 excess returns over 21, 63, 126 and 252 trading days, annualised, in %
- **Moments** (eq. 10): for any nonnegative z_t known at t, $E[(R^e_{t,T} - b_{t,T})\, z_t] \geq 0$; z_t = a constant and nine Welch–Goyal predictors made positive Welch & Goyal (2008) (Table 3)

Case Study: Data and Method (2/2)

- **Kodde–Palm tests** Kodde & Palm (1986) (eqs. 11–14)
  - $\bar{\lambda}$: the 10 sample means; Σ: their Hansen–Hodrick covariance Hansen & Hodrick (1980), lags = days in the horizon
  - $D_1 = \min_{\lambda \geq 0}(\lambda - \bar{\lambda})^\top \Sigma^{-1}(\lambda - \bar{\lambda})$ tests validity (H_0: $\lambda_0 \geq 0$)
  - $D_2 = \bar{\lambda}^\top \Sigma^{-1} \bar{\lambda} - D_1$ tests tightness (H_0: $\lambda_0 = 0$ vs $\lambda_0 \geq 0$); both follow mixtures of χ^2 laws
- **Forecasting**: bias-corrected predictive regressions Amihud & Hurvich (2004) (Tables 5, 6) and monthly out-of-sample forecasts against the historical mean (Table 8; R^2_{OS}: out-of-sample R^2 against the historical mean; protocol: Appendix)

Case Study: Results (Tables 1, 3, 5, 6 and 8)

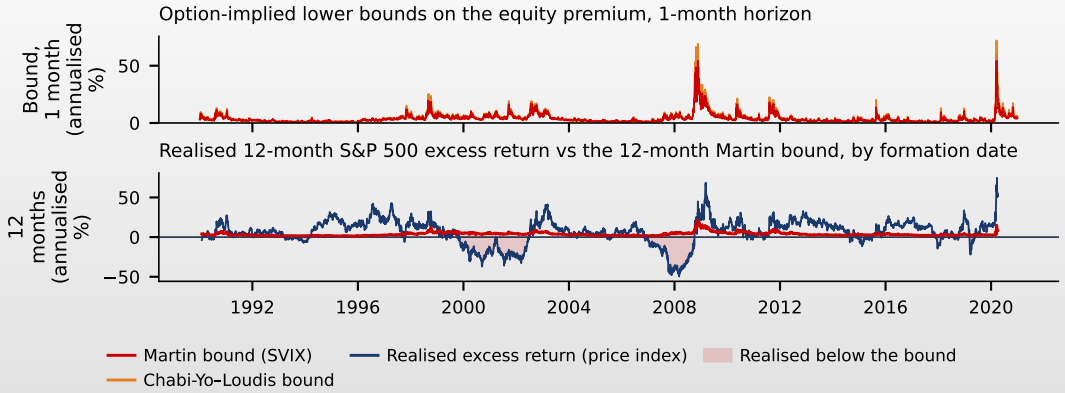
Horizon	Mean R^e (%)	Mean b (%)	D_1	D_2	$\hat{\beta}$ (SE)	R^2_{OS} (%)
1 month	8.76	3.81	0.00	29.93	1.36 (1.23)	−1.2
3 months	8.73	3.86	0.00	39.51	1.39 (1.47)	−1.2
6 months	8.67	3.83	0.00	49.28	1.78 (0.85)	−0.2
12 months	8.86	3.68	0.00	71.21	1.04 (0.88)	−3.0

- Source: Back, Crotty & Kazempour (2022), *Journal of Financial Economics*, Tables 1, 3, 5, 6 and 8
- **Validity not rejected**: all 10 sample moments are positive, so $D_1 = 0$ and $p = 100\%$ at every horizon; this does not prove validity in every state
- **Not tight**: D_2 exceeds 28.86, the upper 0.1% Kodde–Palm critical value for 10 restrictions, at every horizon (finite-sample p: 3.6, 0.4, 2.6, 8.3%); mean slackness 4.8 to 5.2 points a year: the bound is less than half of the realised premium
- **Forecasting**: slopes positive but imprecise (Table 5)
  - With the nine predictors as controls: a one-s.d. rise in the 12-month bound, a 6.6-point higher 12-month excess return (Table 6)
  - Bootstrap p-values 20.7%, below 0.1%, 0.2%, 9.5%; with a Holm correction for the four horizons (Section 11), only 3 and 6 months stay significant at 5%
- **Out of sample**: the Martin bound loses to the historical mean at every horizon; the best case, Chabi-Yo–Loudis at 6 months, reaches 4.3%, not significant (Table 8): low power is one possible explanation

Case Study: Do the Two Tightness Tests Agree?

- Same D_2, two yardsticks: the asymptotic Kodde–Palm critical value (28.86 at 0.1%, 10 restrictions) rejects tightness at every horizon; the finite-sample p-values 3.6, 0.4, 2.6, 8.3% reject at 5% at 1, 3 and 6 months, but not at 12 months
  - Without a multiple-testing correction: not tight at 1–6 months on both tests; at 12 months only the asymptotic test rejects
  - With a Holm correction for four horizons (Section 11), the finite-sample test rejects only at 3 months ($0.4\% < 1.25\%$, then $2.6\% > 1.67\%$); the asymptotic test rejects at every horizon even after the correction
- Why the two can disagree: 1990–2020 holds only about 31 non-overlapping 12-month windows, and the moments use persistent predictors; the HAC covariance Σ is then imprecise and asymptotic critical values can be too small
  - The finite-sample p is the more cautious reading; report both, never only the one that rejects (Seminar 12, C3)
- The mean slackness of 4.8–5.2 points a year does not depend on the test: economically, the bound covers less than half of the realised premium with dividends (Table 1) at every horizon

Case Study: the Bound and the Realised Premium (1/3)



MFM_ch12_bck_bounds

Case Study: the Bound and the Realised Premium (2/3)

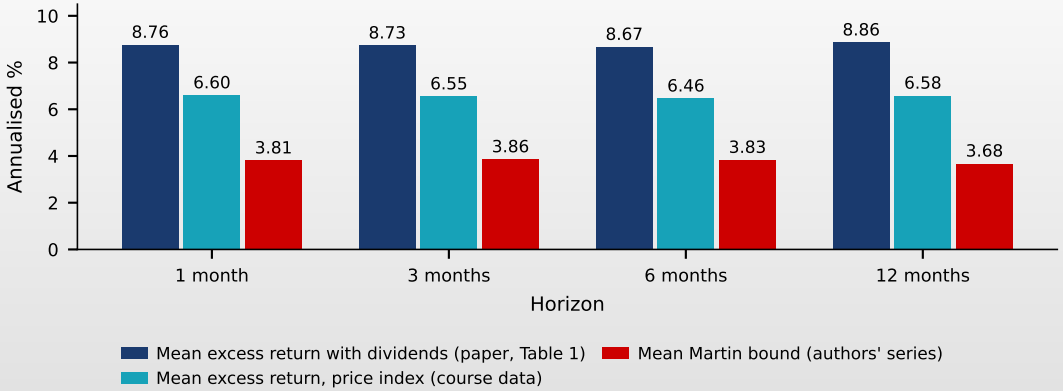
- Top: the authors' daily 1-month bounds, 1990–2020 (replication package)
  - Bottom, course data: 12-month excess return of the S&P 500 price index vs the 12-month Martin bound
- 1-month bound: median 2.7%, peak 54.4% (16 March 2020)
  - 12-month peak 21.1% (20 November 2008): a risk-neutral variance, it jumps with the VIX in crises

Case Study: the Bound and the Realised Premium (3/3)

- Chabi-Yo–Loudis $\geq$ Martin on 99.99% of days
  - Negative skewness and fat tails raise it
- Realised 12-month excess return below the bound on 33% of dates
  - 2000–2002, 2007–2008: the bound restricts $E_t[R^e_{t,T}]$, not each realisation
- Correlation bound–realised 0.15 (overlapping, descriptive)
  - Return noise dwarfs the bound: imprecise slopes, negative R^2_{OS}

MFM_ch12_bck_bounds

Case Study: Mean Premium and Mean Bound (1/2)



MFM_ch12_bck_bounds

Case Study: Mean Premium and Mean Bound (2/2)

- Dark blue: Table 1 of the paper (S&P 500 with dividends); teal: course data (price index, the same formation days: 7789 at 1 month, 7598 at 12 months); red: mean of the authors' Martin bound
- Mean slackness on the course data 2.79, 2.69, 2.63, 2.91 points a year vs 4.8–5.2 in the paper: the gap of 2.2–2.3 points is mostly the missing dividend yield
- The mean bound covers 41–44% of the premium with dividends, 56–59% of the price-index premium; single realisations fall below it on 41% of days at 1 month (12 months: the bound and realised premium slides above), the average never (Seminar 12, C3)

MFM_ch12_bck_bounds

Case Study: Replication, Limitations and an Open Extension (1/3)

- **Replicate** (Seminar 12, C3)
  - Data: the authors' daily bound series (replication package), S&P 500 daily closes of the course data
  - The daily risk-free rate (Kenneth French Data Library); the Welch–Goyal predictors (Amit Goyal's data page)
  - Rebuild Table 1 and D_1, D_2 of Table 3; the course index has no dividends, so excess returns and slackness are lower: a data limitation

Case Study: Replication, Limitations and an Open Extension (2/3)

- **Limitations stated by the authors**
  - A short series, limited to the years after 1990 by the need for option prices: the out-of-sample tests have little power (paper, Section 6)
  - Validity holds even when the bound is replaced by its time-series mean, so it says nothing about predictive power (paper, Section 4.3)
  - More recent bounds and formulas for risk premia are not tested (paper, Introduction)

Case Study: Replication, Limitations and an Open Extension (3/3)

- **Open extension**: the course data have VIX, not SVIX; VIX weights OTM options by $1/K^2$ and exceeds SVIX when the left tail is priced (Martin (2017), Result 3)
  - Is the Martin bound still valid on 2021–2026 (the 2022 bear market, the 0DTE era of Section 9)?
  - Is a VIX-based bound still valid?
  - If not, can the Kodde–Palm test detect it?

The Section's Question and the Tools to Test It

- The section's question: does the VRP (Section 8) still predict equity returns after 2007? Below, the tools to test it
- **Hodrick 1B** Hodrick (1992): instead of regressing overlapping h-month returns on VRP_t, regress the one-month return on the sum of the last h values of the regressor; the overlap moves to the regressor side, and the errors are no longer overlapping
- $R^2_{OS} > 0$: the predictive model has a lower mean squared forecast error than the historical mean; **Clark–West** Clark & West (2007) tests this for nested models, adjusting for the noise that estimating a truly zero slope adds to the larger model
- **Holm correction** for m tests (e.g. four horizons): sort the p-values, compare the smallest with α/m, the next with $\alpha/(m-1)$, and so on; stop at the first that fails
  - Toy check: $p = 0.01, 0.03, 0.04$, $\alpha = 0.05$, $m = 3$: $0.01 < 0.0167$ rejects, $0.03 > 0.025$ stops, so only one hypothesis is rejected, although all three $p < 0.05$

An Open Question: Does the VRP Still Predict Returns?

- Bollerslev, Tauchen & Zhou (2009), Table 2: monthly data 1990–2007, $y_t^{(h)} = \frac{12}{h}\sum_{j=1}^{h}(r_{m,t+j} - r_{f,t+j})$ on $\mathrm{VRP}_t = \mathrm{VIX}_t^2/12 - \mathrm{RV}_t$
  - RV_t: realised variance of month t; at $h = 3$: slope 0.47, Hodrick $t = 2.86$, adjusted $R^2 = 6.82\%$, the peak of a hump-shaped horizon profile
  - Units: $\mathrm{VIX}_t^2/12$ and RV_t are both monthly, not annualised, $\%^2$; in the course replication RV_t is the sum of squared daily log returns (in %) within month t. Section 8 used annualised $\%^2$, a factor 12 larger
- Goyal, Welch & Zafirov (2024): 29 equity-premium predictors proposed after 2008, re-examined with data to 2021: over one third lose in-sample significance, half of the rest fail out of sample
- Open: is VRP predictability a stable price of variance and jump risk, or a property of 1990–2007? The 2008–2026 sample adds two crashes and the 0DTE era
- Formally: $H_0 : \beta_3 = 0$ vs $H_1 : \beta_3 > 0$ in $y_t^{(3)} = \alpha + \beta_3 \mathrm{VRP}_t + u_t$
  - Out of sample: H_0: the VRP forecast does not lower the mean squared prediction error of the historical mean; Clark–West statistic Clark & West (2007)

The Discovery Loop and What the Researcher Must Check (1/2)

- **Literature**: ask an assistant for peer-reviewed tests of VRP predictability since 2009 (DOI: digital object identifier; sample; table)
  - Check: resolve every DOI on Crossref and compare the cited table with the paper, digit by digit
- **Hypotheses**: time-varying risk aversion, jump-risk compensation, or a small-sample artefact of persistent regressors and overlapping returns
  - Check: each must imply something testable for the horizon profile, for 2008–2026 and for non-US markets

The Discovery Loop and What the Researcher Must Check (2/2)

- **Data and code**: build $\text{VIX}^2/12$ and monthly RV from the daily VIX and S&P 500 closes of the course data, reproduce Table 2 ($h = 3$) before extending anything
  - Check timing (VIX at the end of month t, returns from $t + 1$) and match the published means (implied 33.23, realised 14.93)
- **Robustness and critique**: a dated pre-analysis plan (horizons, start date, benchmark, Holm correction); a hostile-referee prompt
  - Check overlap (Hodrick errors Hodrick (1992)), horizon and start-date mining (R^2_{OS} against the start date), the weight of 2008 and 2020; full prompts: Appendix

Mini-Case: Replicate, Then Extend (1/2)

- A typical assistant proposal: "the VRP predicts the quarterly equity premium; extend the 2009 result to today". We test it on the course data
- Replication, February 1990 to September 2007, $h = 3$, daily data: $\hat{\beta}_3 = 0.42$, $t_{\text{Hodrick}} = 2.25$, adjusted $R^2 = 4.6\%$ (published: 0.47, 2.86, 6.82%)
  - Mean implied variance 33.1 is close to 33.23; mean realised 20.6 vs 14.93: daily close-to-close data
- Extension to May 2026 (436 months): $\hat{\beta}_3 = 0.09$, $t_{\text{Hodrick}} = 1.33$, adjusted $R^2 = 1.0\%$; all horizons with four kinds of standard errors: Appendix

Mini-Case: Replicate, Then Extend (2/2)

- Out of sample from January 2008: 221 expanding-window forecasts
  - $R^2_{OS} = 1 - \sum_t (y_t - \hat{y}_t)^2 / \sum_t (y_t - \bar{y}_t)^2 = -22.2\%$ Campbell & Thompson (2008); $\bar{y}_t$: historical mean up to t
  - Clark–West $t = 0.64$ ($p = 0.26$): H_0 not rejected
- Reading: sign and size replicate in-sample; the longer sample does not support stable predictability. Outliers, measurement or a vanished premium: that is the open part

MFM_ch12_ai_discovery

Your Turn: a Project on the VRP After 2008 (1/2)

- Questions
  - Does the VRP predict the equity premium after 2008?
  - Does the Bitcoin premium of Section 8 (DVOL) predict Bitcoin returns?
- Replicate first: Bollerslev, Tauchen & Zhou (2009), Table 2, $h = 3$ (the published numbers of the open question); explain every gap

Your Turn: a Project on the VRP After 2008 (2/2)

- Deliverables
  - A dated pre-analysis plan; the replication table; out-of-sample results for all horizons with a Holm correction
  - Robustness (subsamples, without 2008 and 2020); a referee-style critique and your reply
  - Optional: realised variance from 5-minute returns, only if you have access to intraday data; the course data table has no intraday series
- Log each prompt that shaped a decision in `AI_USE.md`; each invented reference, wrong number or look-ahead caught in `AI_ERRORS.md`, with the test that caught it (templates)

What Option Prices Measure, and Their Limits

- Parity, bounds and static spanning need no model; the tree and the Black–Scholes equation price by dynamic replication
- The Greeks measure exposures; discrete hedging leaves an error of order $1/\sqrt{N}$
- A delta-hedged option is a bet on realised vs implied variance; implied variance exceeded realised variance on average on the S&P 500 and on Bitcoin
- The surface and the risk-neutral density are estimates: impose no-arbitrage constraints and report bands
- The VIX is a strip estimator with jump and truncation biases; as a forecast it needs joint tests, logs and HAC loss comparisons
- Option-implied bounds on the equity premium are valid but not tight in 1990–2020: evidence rests on average slackness, not on forecasting power
  - "Valid" means not rejected; "not tight" holds at every horizon on the asymptotic Kodde–Palm test; on finite-sample tests, at 1–6 months without a multiple-testing correction and only at 3 months with a Holm correction

Check your understanding (1/2)

- Which pricing restrictions follow from no arbitrage alone?
- Which of the restrictions used in the chapter depend on the Black–Scholes assumptions?
- Why can a delta-hedged option lose money even when the model is right?
- What role does the gap between realised and implied variance play in the hedging result?
- Does a fitted smile automatically exclude arbitrage?

Check your understanding (2/2)

- How do you check the butterfly constraint for an SVI fit?
- Does a risk-neutral density forecast the real-world frequencies of returns?
- Why is the standard error $s/\sqrt{n}$ wrong for a mean over overlapping 21-day windows?
- Which standard error replaces it?
- How can two small t-statistics coexist with a large joint Wald statistic?

Open Questions in Option Research (1/2)

- **Frontier**
  - Andersen, Fusari & Todorov (2015): a left-tail factor in index-option surfaces, not spanned by volatility, predicts future equity and variance premia
  - Bali et al. (2023): nonlinear machine learning on option and stock characteristics predicts equity-option returns out of sample
  - Open: jointly estimated, arbitrage-free surfaces with valid confidence bands; the pricing of 0DTE options
- **Next**: Chapter 13, machine learning for financial prediction, with volatility and option-implied features among the inputs

Open Questions in Option Research (2/2)

- **Further reading**
  - Textbooks: Franke, Härdle & Hafner (2019), Härdle, Chen & Overbeck (2017), Čížek, Härdle & Weron (2011), Gatheral (2006)
  - Pricing kernels: Aït-Sahalia & Lo (1998), Grith, Härdle & Park (2013); crypto options: Matic, Packham & Härdle (2023), Alexander & Imeraj (2020)
  - Quantinar: XFG Advanced Methods in Quantitative Finance, Pricing Kernels and Risk Premia, Hedging Cryptocurrency Options
  - Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_12

Proof of Spanning, Moments and SVIX

- Proof for $S_T > F$: $\int_F^{S_T} f''(K)(S_T - K)\,dK = f(S_T) - f(F) - f'(F)(S_T - F)$ by parts; the put integral vanishes; symmetric for $S_T < F$
- Take $e^{-r\tau}E^Q[\cdot]$ with F the forward (a forward contract costs 0): a bond, a forward and a strip of OTM options with weights $f''(K)$ replicate any smooth pay-off
- **Moments**: $f = [\ln(S/F)]^n$, $n = 2, 3, 4$, price the raw risk-neutral moments of the log return; centring and standardising gives volatility, skewness and kurtosis Bakshi, Kapadia & Madan (2003)
- **SVIX** Martin (2017), from $f = S^2$: $\mathrm{SVIX}_t^2 = \dfrac{2}{\tau R_f S_t^2}\left[\displaystyle\int_0^F P\,dK + \int_F^\infty C\,dK\right] = \mathrm{var}_t^Q(R_T/R_f)/\tau$, with R_T, R_f gross returns
  - SVIX^2 gives the Martin lower bound on the equity premium (case study, Section 10)

Back

Why $1/\sqrt{N}$? The Discrete Hedging Error

- Over one interval $\Delta t = T/N$ the hedged book earns $\frac{1}{2}\Gamma S^2\left[\sigma^2\Delta t - (\Delta S/S)^2\right]$: time decay against realised squared return
- Mean 0 under the model to leading order in Δt; $\mathrm{Var}[(\Delta S/S)^2] \approx 2\sigma^4\Delta t^2$ and the terms are approximately uncorrelated, so $\mathrm{Var}(\text{error}) \approx \frac{1}{2}\sigma^4\Delta t\, E\int_0^T \Gamma_t^2 S_t^4\, dt$
- ATM call (derivation): $E[\Gamma_t^2 S_t^4] \approx \dfrac{K^2}{2\pi\sigma^2\sqrt{T^2 - t^2}}$ integrates to $K^2/(4\sigma^2)$, so s.d. $\approx \sigma K\sqrt{T/(8N)} = \sqrt{\pi/4}\,\mathcal{V}\sigma/\sqrt{N}$
  - With vega $\mathcal{V} = 19.72$ per unit
  - The last equality holds exactly for the vega at $d_1 = 0$, $\mathcal{V}_0 = K\sqrt{T}\,\varphi(0)$; here $d_1 = 0.15$ and $\mathcal{V} = \varphi(0.15)/\varphi(0)\,\mathcal{V}_0 = 0.989\,\mathcal{V}_0$, so at $N = 12$ the two forms give 1.02 and 1.01
- Asymptotic theory of the replication error: Bertsimas, Kogan & Lo (2000); it fails with jumps (the error does not vanish as $N \to \infty$), with stochastic volatility (vega risk) and with costs

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The Gamma Moment Behind the $1/\sqrt{N}$ Law

- ATM call, $S_0 = K$, $r = 0$, remaining time $\tau = T - t$: $\Gamma_t S_t^2 = S_t\,\varphi(d_1)/(\sigma\sqrt{\tau}) \approx K\varphi(d_1)/(\sigma\sqrt{\tau})$, with $d_1 \approx X/(\sigma\sqrt{\tau})$ and $X = \ln(S_t/K) \sim N(0, \sigma^2 t)$
- Gaussian integral: $E[\varphi(d_1)^2] = \dfrac{1}{2\pi} E\left[e^{-X^2/(\sigma^2\tau)}\right] = \dfrac{1}{2\pi}\sqrt{\dfrac{\tau}{\tau + 2t}}$, and $\tau + 2t = T + t$
- Hence $E[\Gamma_t^2 S_t^4] \approx \dfrac{K^2}{2\pi\sigma^2}\,\dfrac{1}{\sqrt{(T-t)(T+t)}} = \dfrac{K^2}{2\pi\sigma^2\sqrt{T^2 - t^2}}$, and $\displaystyle\int_0^T \frac{dt}{\sqrt{T^2 - t^2}} = \frac{\pi}{2}$ gives the integral $K^2/(4\sigma^2)$
- Check with the course numbers ($\sigma = 0.2$, $K = 100$, $T = 0.25$, $N = 12$): $\sigma K\sqrt{T/(8N)} = 20\sqrt{0.25/96} = 1.02$, simulated 0.96

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The Moment Formula for the Wings

- Moment formula of Lee (2004): $\beta_R = \limsup_{k\to\infty} w(k)/k \in [0,2]$ and $\tilde{p} = \frac{1}{2\beta_R} + \frac{\beta_R}{8} - \frac{1}{2}$
  - $\tilde{p} = \sup\{p : E^Q[S_T^{1+p}] < \infty\}$: a steeper wing means fatter tails; the left wing mirrors it with negative moments
- Bitcoin: the largest right slope 0.511 gives $\tilde{p} = 0.54$; a formal extrapolation calculation only: this fitted slice fails a butterfly check, so it does not identify a valid risk-neutral moment
- For SVI a right-wing slope below 2 also gives the tail condition $d_+(k) \to -\infty$ of Gatheral & Jacquier (2014), Lemma 2.2

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Estimators of the Risk-Neutral Density

- Fit a smile (SVI) and differentiate the implied call prices twice, as in the lecture
- Kernel regression of C on (K, τ) Aït-Sahalia & Lo (1998); the same under monotonicity and convexity in K Aït-Sahalia & Duarte (2003)
- Smoothness-penalised implied trees Jackwerth & Rubinstein (1996); shape-invariant pricing kernels Grith, Härdle & Park (2013)
- Tails beyond the quotes: SVI wings or an extreme-value tail; either way an assumption, not an estimate

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What the Strip Measures When Prices Jump

- With jumps J: the strip prices $\int \sigma_t^2 dt + 2\sum(e^J - 1 - J)$, but realised variance is $\int \sigma_t^2 dt + \sum J^2$
  - Gap $2E^Q \sum(e^J - 1 - J - J^2/2) \approx E^Q \sum J^3/3$: negative for crashes, so the strip understates expected realised variance Carr & Wu (2009)
- Merton model of Section 6 (one jump a year, mean -8%): strip 17.82 vs $\sqrt{E^Q[\text{QV}]} = 18.00$ vol points, -2.0% in variance (QV: quadratic variation, the realised variance including jumps)

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Truncation and Discretisation of the Bitcoin Index

- Only a finite set of strikes is quoted: truncating the strip biases the index down, a coarse grid biases it up Jiang & Tian (2005)
- The Bitcoin 30-day index from the same two expiries, in three valid variants:
  - from quoted prices: 40.03
  - from SVI prices at the same strikes: 39.99
  - from SVI on a dense grid inside the quoted range: 39.77
- Truncation does not explain the 1.9-point gap to DVOL: the three variants differ by at most 0.27 points; quote selection and methodology remain
- Extending SVI beyond the quotes ($k \in [-4, 4]$) is not a fourth estimate: the left wing of the 17-day expiry has slope $2.30 > 2$ (Lee's bound), so the extrapolated prices do not come from a valid density, and the integral gives 128.47, a value with no economic meaning

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Encompassing and Out-of-Sample Tests

- HAR-RV Corsi (2009): regressors are the realised variance averaged over the last 1, 5 and 22 trading days (daily, weekly, monthly); encompassing regression in logs Busch, Christensen & Nielsen (2011)
  - The target is $RV_{t,t+21}$ over the next 21 days: 22 is a look-back window, 21 the forecast horizon
  - VIX slope 0.90 (0.056); R^2: VIX 0.570, HAR 0.474, both 0.575
- Out of sample from 2000 (6 697 days; coefficients re-estimated every 21 days on targets already observed)
  - QLIKE loss $L = RV/\hat{\sigma}^2 - \ln(RV/\hat{\sigma}^2) - 1$, robust to noise in the RV proxy Patton (2011)
  - Loss ratio VIX^2 / HAR: 0.94, Diebold–Mariano $t = -0.93$ ($p = 0.35$) Diebold & Mariano (1995); VIX corrected by the Mincer–Zarnowitz regression: 0.95 ($p = 0.35$)
- Targets overlap by 20 of 21 days: the Diebold–Mariano variance is HAC with 21 lags; instrumental variables by just-identified generalized method of moments (GMM) with a HAC covariance

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Does the Ex-Ante VRP Predict Returns? Overlapping-Horizon Inference

Horizon	$\hat{b}$	t_{NW}	t_{HH}	$t_{Hodrick}$	p_{boot}	R^2 (%)
$h = 1$	0.07	0.33	0.36	0.41	0.35	0.2
$h = 3$	0.09	0.69	0.67	1.33	0.07	1.2
$h = 6$	0.03	0.49	0.48	0.65	0.26	0.3
$h = 12$	−0.02	−0.64	−0.79	−0.54	0.69	0.2

- Monthly 1990–2026: $\frac{12}{h}\sum_{j=1}^{h} r^e_{t+j} = a + b\,\mathrm{VRP}_t + u_t$, $\mathrm{VRP}_t = VIX_t^2/12 - RV_t$, ex-ante proxy with past RV_t as the forecast; r^e_{t+j}: S&P 500 excess return of month $t+j$, h: horizon in months Bollerslev, Tauchen & Zhou (2009)
- Units: $VIX_t^2/12$ (month-end VIX) and RV_t = sum of squared daily log returns (%) within month t, both monthly %^{2}, not annualised: a factor 12 below the annualised VRP of Section 8

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VRP: Inference with Overlapping Horizons

- If one-month errors are serially uncorrelated, overlap makes u_t an MA($h-1$), a moving average of order $h-1$
  - Newey–West (t_{NW}, h lags), Hansen–Hodrick (t_{HH}) Hansen & Hodrick (1980), Hodrick 1B ($t_{Hodrick}$) Hodrick (1992): under the null its size is closer to the nominal level
  - In a given sample $t_{Hodrick}$ can still exceed t_{NW}: at $h = 3$, 1.33 vs 0.69
  - These bandwidths are choices, so check their sensitivity
- p_{boot}: one-sided, VRP simulated as AR(1), a first-order autoregression ($\hat{\phi} = 0.13$) under $b = 0$, residuals resampled jointly Stambaugh (1999)
- On the full 1990–2026 sample no horizon is significant; on 1990–2007 (Section 11) $h = 3$ is significant

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Case Study, Filters and Forecasting Protocol

- Option data: Cboe for 1990–1995 and 2020–2021, OptionMetrics for 1996–2019
- Filters: standard options only (no weekly or quarterly), bid > 0, the OTM option at each strike, 7 to 550 days, at least 10 strikes per expiry
- Returns: S&P 500 with dividends minus the Fama–French risk-free rate, up to March 2021
- Out of sample (Table 8): monthly forecasts against the expanding post-1926 market mean, 60-month initial window
  - $R^2_{OS} = 1 - \sum \varepsilon_t^2 / \sum \nu_t^2$, ε_t and ν_t the errors of the forecast and of the benchmark; Diebold–Mariano tests Diebold & Mariano (1995)

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The Discovery Loop, Reusable Prompts (1/2)

- **Literature**: "List peer-reviewed tests of VRP return predictability since 2009: DOI, sample, horizon, table, headline estimate."
  - Search with Semantic Scholar, Google Scholar, Elicit; checks: main slide
- **Hypotheses**: "Give three competing explanations of VRP predictability and the testable implication of each for the horizon profile, for 2008-2026 and for non-US markets."
- **Data and code**: "Write Python that builds VIX squared/12 and monthly realised variance from the daily VIX and S&P 500 closes of the course data and reproduces Table 2 of Bollerslev, Tauchen and Zhou (2009), h = 3, with Hodrick (1992) standard errors."
  - Any assistant (Claude, ChatGPT, Gemini, GitHub Copilot)

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The Discovery Loop, Reusable Prompts (2/2)

- **Robustness**, fixed before the results: `"Draft a pre-analysis plan: horizons, out-of-sample start date, benchmark, test statistic and the multiple-testing correction across horizons."`
  - Placebo: VRP lagged 12 months or block-shuffled; subsamples; 5-minute realised variance of SPY (from 2020) instead of daily squared returns (optional: needs intraday data, not in the course data)
- **Critique**: `"Act as a hostile referee: why could this out-of-sample R2 be spurious or biased, and which test settles each objection?"`
  - Expect: the checks of the main slide, plus the bias with persistent regressors
- **Report**: every horizon and specification tried, in one table; the plan dated before the results; AI use logged

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Acronyms used in this material (1/2)

- **AI**: Artificial Intelligence
- **AR**: AutoRegressive model
- **ATM**: At-The-Money: strike close to the forward, $k = \ln(K/F) \approx 0$
- **BTC**: bitcoin, currency code of the cryptocurrency
- **COVID-19**: COronaVIrus Disease 2019
- **CRR**: Cox–Ross–Rubinstein (binomial tree)
- **DVOL**: Deribit Volatility Index
- **ETF**: Exchange-Traded Fund
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GBM**: Geometric Brownian Motion
- **GJR-GARCH**: Glosten–Jagannathan–Runkle GARCH
- **GMM**: Generalized Method of Moments
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HAR**: Heterogeneous AutoRegressive (model)
- **HAR-RV**: Heterogeneous AutoRegressive model of Realised Volatility
- **ITM**: In-The-Money: a put with $K > F$ or a call with $K < F$
- **IV**: Implied Volatility (mark IV; label of the Bitcoin term-structure chart), not instrumental variables
- **MA**: Moving Average

Acronyms used in this material (2/2)

- **OLS**: Ordinary Least Squares
- **OTM**: Out-of-The-Money: a put with $K < F$ or a call with $K > F$
- **QLIKE**: Quasi-LIKElihood (loss function)
- **QV**: Quadratic Variation
- **RV**: Realised Variance
- **SE**: Standard Error
- **SSVI**: Surface SVI (Gatheral–Jacquier parameterisation of the whole surface)
- **SVI**: Stochastic Volatility Inspired (parameterisation)
- **SVIX**: Simple VIX: index of simple variance swaps (Martin, 2017)
- **US**: United States
- **USD**: US dollar, international currency code
- **UTC**: Coordinated Universal Time (Temps Universel Coordonné)
- **VIX**: Cboe Volatility Index
- **VIX3M**: Cboe 3-Month Volatility Index
- **VIX9D**: Cboe 9-Day Volatility Index
- **VRP**: Variance Risk Premium
- **VVIX**: Cboe VIX of VIX Index
- **XFG**: Quantlet prefix of Applied Quantitative Finance (Härdle, Chen & Overbeck)

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